



When Temporal Knowledge Graph meets multi-modality: A new perspective for Temporal Knowledge Graph Forecasting[☆]

Gaojie Han, Wei Chen^{*}, Xiaofang Zhang, Li Zhang^{id}, An Liu, Lei Zhao^{*}

School of Computer Science and Technology, Soochow University, No. 333 Ganjiang Road East, Suzhou, 215006, Jiangsu, China

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ABSTRACT

Temporal knowledge graphs (TKGs) store relational triplets with temporal information, which have attracted increasing attention in numerous downstream applications. Despite the substantial progress made by existing studies on TKG forecasting, they primarily investigate the task on uni-modal TKGs, whereas many knowledge graphs are dynamic and multi-modal. To fill the gap, we extend the TKG forecasting task to embrace multi-modality, yet the following problems pose significant obstacles to our study. (1) How to identify query-related facts from local structural and global temporal perspectives? (2) How to leverage the multi-modal information to distinguish between similar entities (i.e., entities with similar graph information)? To tackle these problems effectively, a novel model namely Multi-modal Temporal Knowledge Graph Forecasting (MM-TKGF) is proposed, which is composed of the following two components. (1) A well-designed representation module is developed to capture entities' temporal dynamics, and learn their temporal representations. (2) A gated multi-modal fusion module is developed to integrate temporal representations of entities with the multi-modal features. The effectiveness of MM-TKGF is assessed across three MMTKG datasets via link prediction, with experimental results demonstrating its superior performance, surpassing existing uni-modal TKG forecasting models by an average of 1.49%, 7.36%, and 6.98% in MRR value.

1. Introduction

Knowledge graphs (KGs) are structured knowledge bases that store, organize, and manage extensive factual information, driving significant advancements in numerous downstream applications, including information retrieval (Wang, Wang, et al., 2023), recommendations (Ren et al., 2024; Zou et al., 2022), social network analysis (Zhan et al., 2021), question answering (Cui et al., 2017; Zhang et al., 2025), and web search (Paulheim, 2017). Traditional KGs are static and typically store uni-modal facts of the form (subject entity, relation, object entity). However, it is important to recognize that factual information constantly changes over time in the real world. Fig. 1 demonstrates the dynamic nature of real-world facts through a sequence of KG snapshots. A pertinent example is the fact (*The United States, Consult, George Bush Sr.*), which maintains validity only in 2014-03, and becomes invalid by 2014-05. To model such entity evolution, temporal knowledge graph (TKG) is formalized as a series of static KG snapshots (Wu et al., 2021; Zhu et al., 2021). Distinct from traditional KGs that organize facts as triplets, TKG facts adopt quadruple representations that explicitly encode temporal information, e.g. (*Barack Obama, Consult, India, 2014-07*). Recently, several large-scale TKGs including

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^{*} Corresponding authors.

E-mail addresses: gjhan@stu.suda.edu.cn (G. Han), robertchen@suda.edu.cn (W. Chen), xzfzhang@suda.edu.cn (X. Zhang), zhangliml@suda.edu.cn (L. Zhang), anliu@suda.edu.cn (A. Liu), zhaol@suda.edu.cn (L. Zhao).

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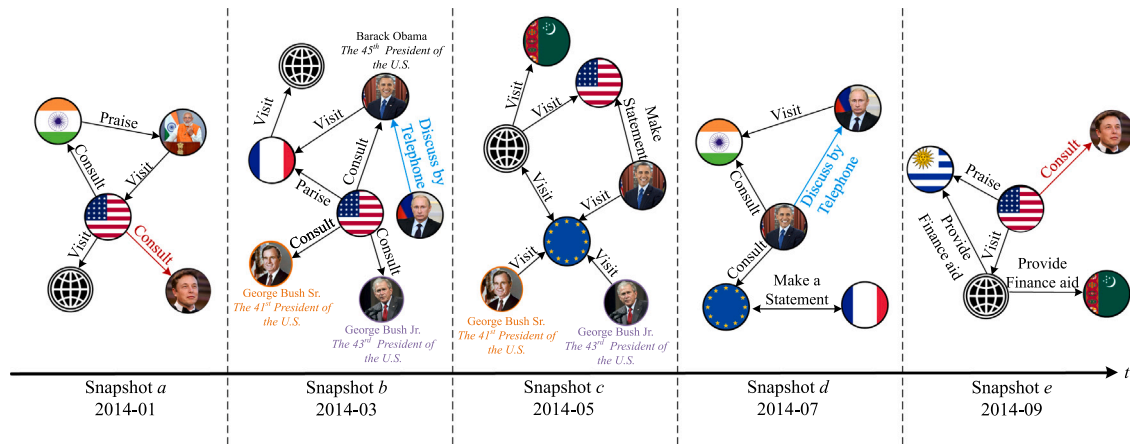


Fig. 1. An example of a series of MMTKG snapshots. The red arrows connect entity *Elon Musk* and its connected coarse-grained entity *The United States*, illustrating the **Problem I** from the structural view. Similarly, the blue arrows highlight the fact (*Vladimir Putin*, *Discuss by telephone*, *Barack Obama*) at various timestamps, demonstrating the **Problem I** from the temporal view. Additionally, the coloured circles, i.e., the entities *George Bush Sr.* (orange circle) and *George Bush Jr.* (purple circle) that can be easily distinguished after adding the multi-modal auxiliary information, serving to illustrate the **Problem II**. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ICEWS¹ and GDELT² have been proposed, offering various temporal facts that facilitate TKG forecasting task (Han, Chen, et al., 2021; Han, Ding, et al., 2021; Xu et al., 2023).

In recent advances, substantial efforts have been made to address the TKG reasoning task, which can be broadly categorized into two primary settings: interpolation (García-Durán et al., 2018; Goel et al., 2020; Lacroix et al., 2020) and extrapolation (Bai et al., 2025; Han, Chen, et al., 2021; Han, Ding, et al., 2021; Liu et al., 2022; Sun et al., 2021; Xu et al., 2023; Zhu et al., 2021). Specifically, interpolation aims to complete missing facts within observed timestamps to enrich TKG, whereas extrapolation focuses on inferring facts that may occur in the future by leveraging historical observations. The main difference between them is that the former focuses on capturing the characteristics of entities in the past, while forecasting in TKG needs to model the evolutionary trend of entities for future prediction. Although promising results have been achieved by these studies, the key idea behind them remains constrained to **infer new knowledge exclusively relying on the graph structure without considering multi-modal data**. This limitation becomes particularly problematic when similar entities share near-identical structural patterns across snapshots. For example, as illustrated in Fig. 1, entities *George Bush Jr.* and *George Bush Sr.* share the same graph structure in Snapshots *b* and *c*. Existing uni-modal methods inevitably embed such entities into similar representations, yielding ambiguous predictions for queries like (*the United States*, *Consult*, ?, 2014-09) and failing to distinguish them accurately. Such phenomena motivate us to leverage multi-modal data to enhance entity representation, as previous multi-modal knowledge graph (MMKG) studies that have proven that traditional KG reasoning methods can benefit from this data (Xie et al., 2019, 2016, 2017; Zheng et al., 2023). As a result, we extend the task of TKG forecasting to encompass multi-modality, and formulate a novel task, i.e., multi-modal temporal knowledge graph (MMTKG) forecasting. Technically, a straightforward method to achieve the extension is to adopt or follow the solutions proposed by existing MMKG work (Liu et al., 2023; Zhang, Chen, Liang, et al., 2024), i.e., embedding entities with all multi-modal information at early stage, then implement downstream tasks based on entity representations. However, the strategy is inappropriate for our study. This is because, different from MMKGs that store all static data in a single graph, the data in MMTKGs is dynamic and distributed across different snapshots. In this case, embedding entities with all data at early stage will lead to the model being biased towards multi-modal data, which contains more features than the structural and temporal data in each snapshot. In fact, the structural and temporal data play a primary role in forecasting model, while multi-modal features act as auxiliary information to refine entity representation, and the experiments in Section 5.4 also illustrate this. Recognizing the shortcomings of existing methods in our study, we prioritize extracting the structural and temporal information of entities first, then incorporate multi-modal features by integrating them with a dynamic weight mechanism (i.e., intermediate fusion). This strategy achieves dynamic equilibrium between different types of information, enabling fine-grained multi-modal feature fusion within our model. Nevertheless, while this strategy brings significant benefits, it also presents the following problems for high-performance MMTKG forecasting.

Problem I (*How to Effectively Identify Query-related Facts from Local Structural and Global Temporal Perspectives?*). (1) From the local structural view, the performance of MMTKG forecasting can be enhanced by retrieving facts related to the query within each snapshot. Actually, the fine-grained entities are frequently linked with query-irrelevant entities that are coarse-grained and

¹ <https://dataverse.harvard.edu/dataverse/icews/>

² <https://www.gdeltproject.org/>

indiscriminate, such as concepts, nations, and organizations. Fig. 1 presents an example that the fine-grained entity *Elon Musk* connects to the coarse-grained entity like *The United States* (indicated by red arrows), forming the common sense fact such as (*The United States, Consult, Elon Musk*). While such facts may offer essential structural information about entities, their extensive presence across various snapshots results in the redundancy of temporal facts, as they are common sense and lack distinctiveness. As a result, it becomes essential to design a strategy to prune these redundant facts out during training and emphasize the importance of query-related facts in the local snapshot. (2) From the global temporal perspective, searching query-related facts across snapshots is equally crucial for enhancing MMTKG forecasting. In Fig. 1, we present an example (marked by the blue arrows) using the temporal fact (*Vladimir Putin, Discuss by telephone, Barack Obama*) at various timestamps to illustrate this process. In reality, the telephone calls between the presidents of the United States and the Russian Federation are a routine diplomacy event, meaning that the corresponding facts only appear in specific snapshots. Thus, for the query (*Vladimir Putin, Discuss by telephone, ?, 2014-11-01*), the query-related facts such as (*Vladimir Putin, Discuss by telephone, Barack Obama*) appear exclusively in Snapshots *b* and *d*, while temporal facts from Snapshots *a*, *c*, and *e* are deemed semantically irrelevant to the query. Here, the temporal facts in Snapshots *b* and *d* are more critical as they provide evidence to support the prediction results, so emphasizing the importance of them can improve the reasoning model's performance. The aforementioned example demonstrates that effective forecasting requires models to identify important temporal facts (i.e., highly pertinent ones) as evidence and differentiate them from irrelevant facts. Regrettably, existing studies lean towards prioritizing events that are frequent and recent as predicted candidates, ignoring the importance of query-relevant temporal facts, significantly limiting the model's ability to forecast the fact that may occur in the future (Zhu et al., 2021). Therefore, how to identify query-related facts from both structural and temporal perspectives remains a non-trivial task.

Problem II (*How to Leverage Multi-modal Information to Distinguish between Similar Entities?*). Existing TKGs contain only structural and temporal data in the form of facts. However, it is difficult to distinguish entities with similar behaviours relying solely on structural and temporal information, such as the descendants from eminent families. Fig. 1 illustrates this problem through the Bush family (denoted by the coloured circle), where the father entity *George Bush Sr.* (orange circle) and son entity *George Bush Jr.* (purple circle) share most of the fundamental structural information and are former U.S. presidents, exhibiting similar behaviours within the same MMTKG snapshot. Without the auxiliary information, distinguishing these entities is a difficult task. Furthermore, inheriting the name of father is a deeply ingrained tradition in many cultures around the world, and the widespread existence of this phenomenon makes the matter worse. To address the issue, we posit that no two entities can be exactly the same, i.e., even if similar entities share the same graph structure, they can be distinguished with more diverse auxiliary information. Continuing the example from Fig. 1, if the multi-modal auxiliary data is integrated into the entity *George Bush Sr.* (with his photograph and textual description *The 41st President of the United States*), we can clearly distinguish it from the entity *George Bush Jr.* (with his photograph and textual description *The 43rd President of the United States*). The diverse auxiliary data can refine the entity representations, thereby enhancing the reasoning model's performance. However, current MMKG reasoning models only conform to the static scenarios, they cannot simultaneously exploit entity multi-modal and temporal information, which limits their robustness and generalization. Thus, how to exploit multi-modal information to distinguish between similar entities remains an open problem in the TKG reasoning area.

To resolve the above problems effectively, a brand new model namely **Multi-modal Temporal Knowledge Graph Forecasting** (MM-TKGF) is proposed, which comprises two core modules. (1) To tackle the Problem I, we propose a multi-faceted entity representation module called **Attention-based Multi-faceted Encoder (AMFE)**. This module contains the following two independent components. Specifically, to model structural dependencies within MMTKG snapshots, a structural encoder that comprises a sequence of **Attentive Graph Neighbour Encoder (AGNE)** is proposed. Notably, AGNE utilizes relational graph convolutional network coupled with query-dependent attention mechanism to dynamically aggregate the most important structural information from the structural perspective, and generate entity and relation structural representations. Additionally, to capture the temporal dynamics within MMTKG, we design a temporal encoder that comprises a series of **Attention-based Temporal Encoders (ATE)**. Especially, ATE leverages self-attention mechanism to highlight valuable query-related temporal dynamics from the temporal perspective, and refine entity temporal representations. Through these two components, the module AMFE can effectively identify query-related facts from both local structural and global temporal views. (2) To address the Problem II, a multi-modal fusion module, entitled **Temporal-aware Gated Multi-modal Fusion Module (TGMF)**, is proposed. This module enables the model to effectively deal with inter-modal interactions and fuse multi-modal features with entity temporal representations. Particularly, the gated fusion unit in TGMF further enables it to filter the inter-modal noise out, thereby generating a better unified representation for an entity. Finally, the model can differentiate between similar entities effectively, and predict the fact that may occur in the future.

In short, we summarize the primary contributions of the paper as follows:

- To our best knowledge, we are the first to extend the scope of TKG forecasting into a multi-modal scenario, and formulate the task of MMTKG forecasting.
- A brand new model MM-TKGF for MMTKG forecasting is proposed, where the first module AMFE is designed to address the Problem I, by leveraging a relational graph convolutional network coupled with attention mechanism to capture the evolutionary patterns of entities. The second module TGMF is developed to solve the Problem II, by designing a temporal-aware gating network to integrate temporal representations with multi-modal features adeptly.
- To fully investigate the effectiveness of MM-TKGF, exhaustive experiments are performed across three MMTKG datasets (i.e., MMICEWS'14, MMICEWS'18, and MMICEWS'05-15), with the results indicating the superiority of MM-TKGF over both uni-modal and multi-modal KG reasoning baselines.

Compared to the conference version of this work (Han et al., 2023), the present paper has the following advancements:

- Apart from the graph structural and temporal information utilized in (Han et al., 2023), the multi-modal data is further exploited to enhance the entity representation, thereby improving the model's performance. The traditional TKG forecasting task is extended to a more complex setting, requiring the model to have the ability to merge temporal and multi-modal features.
- We optimize the previous uni-modal TKG forecasting model developed in (Han et al., 2023). A carefully designed gated fusion module is proposed to incorporate the temporal with multi-modal representations of entities, thereby achieving accurate MMTKG forecasting.
- We construct three novel MMTKG datasets, denoted as MMICEWS'14, MMICEWS'18, and MMICEWS'05-15, where the original ICEWS data as the backbone for the constructed MMTKGs, textual and image information as supplementary. These datasets offer a new perspective for benchmarking TKG reasoning and forecasting models, which provide valuable insights for the community.

The rest of this paper is outlined below. Section 2 conducts a thorough review of foundational work in TKG and MMTKG reasoning. Section 3 defines important notions and formulates the MMTKG forecasting problem. The details of MM-TKGF, including its AMFE and TGMF modules, are systematically elaborated in Section 4, followed by Section 5 evaluating the model through extensive experiments across three MMTKG datasets. Section 6 then discusses the theoretical and practical implications of our findings. Finally, we summarize the work and explore promising avenues for further investigation in Section 7.

2. Related work

In this section, we commence by reviewing the foundational work of TKG reasoning, and then perform a brief review of MMKG with its reasoning methods.

2.1. Temporal knowledge graph reasoning

Reasoning is an extremely useful tool in completing and forecasting facts in TKG area, where incorporating temporal information is a crucial problem. To address the issue, a considerable number of researchers have shared their ideas from the following two perspectives: interpolation (Sadeghian et al., 2021; Wang et al., 2014) and extrapolation (Han, Chen, et al., 2021; Liu et al., 2022).

Interpolation Methods. These methods aim to enrich incomplete TKGs by completing the missing facts at historical timestamps. A classic solution is to generalize existing methods for static KG embedding methods, enabling the model to incorporate temporal information (Dasgupta et al., 2018; Goel et al., 2020). For example, TTransE (Jiang et al., 2016), which extends the classic embedding method TransE (Bordes et al., 2013) by incorporating temporal-based representation into the corresponding entity and relation embeddings. Similarly to TTransE, Dasgupta et al. (2018) is an extension for the translation-based model TransH (Wang et al., 2014), which projects entities and relations to the time-specific hyper-planes (Wang et al., 2014). Moreover, Goel et al. (2020) leverages diachronic word embedding to generate the temporal-specific entity representations and uses the Simple scoring function (Kazemi & Poole, 2018) to measure the quadruple plausibility. García-Durán et al. (2018) adopts a sequence model to incorporate temporal information into relation representation while preserving the scoring function used in DistMult (Yang et al., 2014). Furthermore, Ma et al. (2019) proposes a light-weight Tucker tensor decomposition method that learns a time-specific tensor to complete historical missing facts. ChronoR (Sadeghian et al., 2021), an extension of the rotation-based method RotatE, represents the temporal dynamics of entities through scaling and rotation operations in the embedding space. To capture fine-grained temporal dynamics in TKGs, RTGE (Xu et al., 2022) extends TransH by combining a measure of temporal smoothness to generate the robust temporal embedding of entities. While the aforementioned models can predict the missing facts at an observed timestamp to enrich the incomplete TKG, their ability to predict future temporal facts is limited, since they cannot effectively leverage the rich temporal dynamics involved by TKGs.

Extrapolation Methods. Different from interpolation approaches, the extrapolation methods are devoted to utilize past historical observations to infer the facts that may occur in the future (Han, Ding, et al., 2021). Intuitively, accurate forecasting over TKGs requires a model to utilize and capture temporal dependencies effectively. An effective solution to achieve this is to treat TKG as a snapshot sequence, extracting temporal patterns across these snapshots by traversing the timeline from past to present. For instance, GCRN (Seo et al., 2018) introduces an architecture that integrates graph neural network layer with recurrent neural network, modelling TKG as sequential snapshots to jointly capture valuable structural dependencies and temporal dynamics. CyGNet (Zhu et al., 2021) introduces a novel architecture with the copy-generate mode that combines historical and global prediction results to achieve accurate forecasting. Jin et al. (2020) designs a model namely RE-NET that jointly captures structural and temporal dependencies by combining an R-GCN-based neighbour encoder (Schlichtkrull et al., 2018) with an RNN module to capture the temporal dynamics of entities in TKGs. Han, Chen, et al. (2021) introduces an interpretable reasoning method that treats the TKG as a whole, and utilizes a graph attention layer to identify relevant entities within the sub-graph. CEN (Li, Guan, et al., 2022) applies a length-aware convolutional network that dynamically adapts to snapshot sequences with varying lengths, thereby robustly modelling the temporal dynamics of entities. Li, Sun, and Zhao (2022) leverages local recurrent and global history encoders to model sequential, repetitive, and cyclical patterns of historical observations. Li et al. (2024a) introduces two latent variables to jointly capture the dynamic and static characteristics of entities in a unified way. Moreover, a symbolic method TLogic (Liu et al., 2022) is specifically designed for temporal reasoning task. The model extracts temporal logical rules from historical facts, applies them to infer answers,

and then ranks the candidates by evaluating the confidence of temporal-aware rules. HFL (Xu et al., 2025) innovatively transforms TKG reasoning into the sequence prediction task, and leverages pre-trained language models to model periodic temporal information within TKG. Re-Temp (Wang, Han, & Poon, 2023) designs a skip information flow mechanism with explicit temporal embedding to dynamically filter unnecessary information for prediction. Zhang, Sun, et al. (2024) proposes a dual-encoder coupled with a gate mechanism that effectively leverages historical and potential features of TKGs. LogCL (Chen et al., 2024) employs a local-global history-aware framework with entity-aware attention and contrastive learning to enhance noise resistance for extrapolation. However, the aforementioned models often struggle to forecast unseen entities in TKGs. To overcome this, Sun et al. (2021) presents a path-based approach that combines a reinforcement learning model coupled with a mean inductive method, enabling the reasoning of unseen entities over TKGs by leveraging historical paths and temporal patterns. Fan et al. (2025) proposes a sub-graph model using candidate-oriented relational graphs with directed temporal sampling to capture complex relation patterns in inductive reasoning while providing interpretable evidence. Furthermore, Meng et al. (2024) and Bai et al. (2025) address the hard case of reasoning over sparse TKG, and utilize a path embedding model coupled with a soft-reward mechanism to alleviate the sparsity of reasoning paths.

Overall, upon a comprehensive review of existing work, we have observed that no prior studies have applied multi-modal auxiliary information to enhance TKG reasoning. The research most closely related to our work includes the study which leverages event KG with textual descriptions to enhance multi-event forecasting (Deng et al., 2020), as well as the study that utilizes multi-modal temporal knowledge graphs to enhance POI recommendation performance (Liu et al., 2023). Despite the substantial contributions made by prior work, they have not explored the MMTKG forecasting problem so far, thus our proposed model MM-TKGF can fill this gap.

2.2. Multi-modal knowledge graph reasoning

Traditional KGs contain only graph structural information, which are composed of relations, entities and facts (Li, Chen, et al., 2025; Li, Huang, et al., 2025). Currently, the Internet data is inherently multi-modal in nature (Ji et al., 2022). This trend highlights the necessity for more comprehensive data integration, leading to the emergence of MMKGs. Specifically, MMKGs combine the graph structure backbone of KGs (i.e., triplets) with multi-modal auxiliary information (e.g., texts, images) to enrich entity representations (Ferrada et al., 2017; Liu et al., 2019).

Multi-modal Knowledge Graph. The development of MMKGs is accompanied by an increase in the diversity and volume of multi-modal data. The early MMKGs are limited to a single auxiliary modality. For instance, (Ferrada et al., 2017) introduces a novel dataset namely IMGpedia, which uses entities from DBpedia as its backbone, with multi-modal auxiliary data exclusively consisting of images. Similarly to IMGpedia, Liu et al. (2019) and Pezeshkpour et al. (2018) extend static KGs such as WN-9, FB15K, and YAGO-10 to multi-modal scenarios using the same approach. In addition to incorporating image data, utilizing textual information as a multi-modal data source is also a viable option. For example, FB-Des (Xie et al., 2016) takes FB-15k237 as its backbone, and integrates the graph structure with textual description of each associated entity. Although these efforts introduce a clear methodology to construct an MMKG, they often overlook the quality of multi-modal data while enriching entities. To tackle this issue, Richpedia (Wang et al., 2020) designs a retrieval model that filters abnormal images and ensures that the images are highly relevant to the entities. Compared to incorporating single auxiliary modality data, the integration of diverse multi-modal information can provide entities with richer complementary features. Therefore, two MMKGs that contain multiple data modalities (i.e., image and textual information), namely WN9-IMG-TXT and FB-IMG-TXT (Mousselly-Sergieh et al., 2018) are proposed. However, they still assume the entity is static rather than dynamic, which limits the ability of existing MMKG reasoning models to represent entities and relations effectively.

Multi-modal Fusion Strategies. Despite the multi-modal information being able to provide entities with richer complementary features, effectively utilizing this information poses significant challenges. For example, while different modalities provide abundant information, they also introduce significant inter-modal and intra-modal noise. The noise is mainly contained in redundant and irrelevant multi-modal features, which strictly harms the quality of entity representation. Thus, designing an effective strategy to filter out noise and align features from different modalities becomes imperative. Early multi-modal learning methods, as pioneers in fusion strategies, often fuse the feature vectors from different modalities through a direct concatenation or addition operation (Chen et al., 2022; Zheng et al., 2023). As a result, these approaches usually have a major limitation, i.e., simple concatenation or addition cannot filter out the multi-modal noise. To address the problem, some studies apply self-adaptive weight mechanisms, such as attention or gating mechanisms, to extract important features from auxiliary modalities. For instance, MFB/MFH (Yu et al., 2018) applies an attention mechanism to aggregate essential information across all modalities into a fused representation. While attention-based fusion strategies extend to aggregate essential features across all modalities, coarse-grained fusion may result in the fused representation still containing noise. To alleviate this problem, MCAN (Yu et al., 2019) adopts a self-attention mechanism to dynamically fuse multi-modal features in a fine-grained manner. Notably, some MMKG studies also design their unique fusion strategies by using a self-adaptive weight mechanism. For example, MMKGR (Zheng et al., 2023) designs a gated-attention mechanism to filter out inter-modal and intra-modal noise to extract the entity fine-grained features. (Zhang, Chen, et al., 2024) and Jian, Zhang et al. (2025) respectively address imbalanced modality distributions in MMKG via a relation-guided fusion module and an adaptive modality interaction transformer. Although the aforementioned methods have developed effective solutions to fuse features from different modalities, they assume entities are static rather than dynamic. In other words, they cannot integrate the static multi-modal features with dynamic entity temporal representations. This limitation motivates us to design a temporal-aware fusion strategy.

Table 1
Key notations with their descriptions.

Notations	Descriptions
$\mathcal{G}$	An MMTKG
$\mathcal{G}_t$	A snapshot of MMTKG at timestamp t
$\mathcal{E}$	An MMTKG's entity set
$\mathcal{E}_t$	An MMTKG's entity set at timestamp t
$\mathcal{R}$	An MMTKG's relation set
$\mathcal{R}_t$	An MMTKG's relation set at timestamp t
$\mathcal{F}$	An MMTKG's fact set
$\mathcal{F}_t$	An MMTKG's fact set at timestamp t
$\mathcal{M}_e$	An MMTKG's multi-modal feature vector set
$\mathcal{M}_e^t$	An MMTKG's textual feature vector set
$\mathcal{M}_e^i$	An MMTKG's image feature vector set
$\tilde{x}_{e,t}$	Learnable structural embedding of entity e at timestamp t
$\tilde{h}_{e,t}$	Learnable temporal embedding of entity e at timestamp t
$\tilde{m}_e$	Learnable multi-modal embedding of entity e at timestamp t
$\tilde{m}_e^t$	Pre-trained textual embedding of entity e
$\tilde{m}_e^i$	Pre-trained image embedding of entity e
$\tilde{z}_{e,t}$	Learnable unified embedding of entity e at timestamp t
m	Observation time window(s)
d	Embedding dimension(s)
d_t	Pre-trained textual embedding dimension(s)
d_i	Pre-trained image embedding dimension(s)

Static Multi-modal Knowledge Graph Reasoning. Existing MMKG reasoning methods focus on employing various multi-modal fusion strategies to extend KG reasoning into multi-modal scenarios. The classic work TransAE, an extension of translation-based methods, demonstrates that multi-modal auxiliary information can benefit traditional KG reasoning approaches (Wang et al., 2019). IKRL (Xie et al., 2019) leverages image data as multi-modal auxiliary information to improve reasoning accuracy. MKRL (Tang et al., 2019) utilizes textual data as multi-modal auxiliary information to enhance reasoning performance. Chen et al. (2022) introduces a transformer-based architecture to simultaneously capture important features from textual and image data. Similarly, SGMPT (Liang et al., 2024) leverages a multi-modal pre-trained transformer that explicitly integrates structural information to guide multi-modal feature encoding. SNAG (Chen et al., 2025) proposes a unified transformer-based model with modality-level noise masking to embed entities within MMKG through multiple tasks. Moreover, APKGC (Jian, Luo, et al., 2025) designs an attention penalty mechanism coupled with an adaptive noise sampling strategy to address multi-modal illusion in pre-trained models, thereby enhancing completion performance. In a word, the above efforts have introduced various fusion strategies to better utilize the multi-modal auxiliary data in MMKG. Other studies are mainly aiming to address hard cases or complex application scenarios, such as multi-hop reasoning (Zheng et al., 2023), incremental learning (Wei et al., 2023), and few-shot learning (Wei et al., 2024). Although we have witnessed the flourishing development of MMKG reasoning, research on MMTKG forecasting remains unexplored, which motivates us to propose an effective solution for this novel task.

3. Preliminary and definition

In this section, key notations and definitions employed in the paper are introduced first, followed by a problem formulation for MMTKG forecasting. Notably, to better understand the details of MM-TKGF, Table 1 summarizes key notations with their descriptions.

3.1. Key notations and definitions

Definition I (Multi-modal Temporal Knowledge Graph). Existing TKGs typically contain only graph structural information, we address this limitation by introducing a new form of TKG that integrates multi-modal auxiliary data with the fundamental graph structure. To be specific, as the extension of traditional TKG, an MMTKG, which is denoted as $\mathcal{G} = \{\mathcal{E}, \mathcal{R}, \mathcal{F}\}$, can be formalized as a snapshot sequence $\{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_t, \dots, \mathcal{G}_n\}$. Each snapshot $\mathcal{G}_t = \{\mathcal{E}_t, \mathcal{R}_t, \mathcal{F}_t\}$ is a directed heterogeneous graph at timestamp t , where $\mathcal{E}_t$ denotes entity set associated with multi-modal features, $\mathcal{R}_t$ is relation set, and $\mathcal{F}_t$ represents fact set, at timestamp t . The form of multi-modal temporal fact is represented as a quadruple (e_i, r, e_j, t) , where $e_i, e_j \in \mathcal{E}$, and $r \in \mathcal{R}$, indicating that a subject entity e_i connects to an object entity e_j through a relation r at timestamp t .

Definition II (Multi-modal Auxiliary Feature).

The multi-modal auxiliary features of an MMTKG are a series of pre-trained characteristics that are extracted from various entity properties, such as textual descriptions, literal values, geographical coordinates, or images from multimedia sources. For each entity e , its multi-modal feature is represented as m_e , which is then expressed as a d -dimensional vector $\tilde{m}_e = f(\tilde{m}_e^t, \tilde{m}_e^i)$. To be specific,

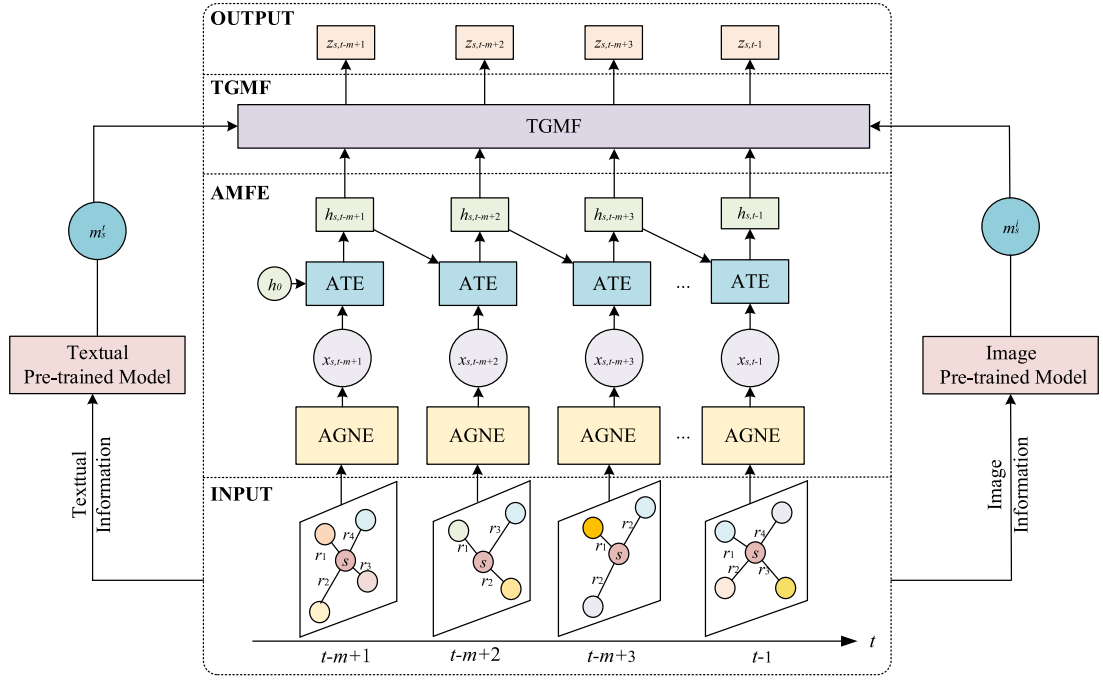


Fig. 2. The overall architecture of MM-TKGF.

$\bar{m}_e \in \mathcal{M}_e$ is learned from two types of multi-modal data (i.e., text and image), where $\bar{m}_e^t \in \mathbb{R}^{d_t}$ represents the textual feature vector, $\bar{m}_e^i \in \mathbb{R}^{d_i}$ denotes the image feature vector, $f(\cdot)$ refers to a multi-modal fusion method, and $\mathcal{M}_e = \{\bar{m}_e | e \in \mathcal{E}\}$ is a set of multi-modal feature vectors for all entities.

3.2. Problem formulation

The paper focuses on inferring the missing subject s or object entity o from a corrupted quadruple based on historical snapshots $\mathcal{G} = \{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_{t-1}\}$, and formally formulates the problem as follows.

Problem (Multi-modal Temporal Knowledge Graph Forecasting). Given a query at a future timestamp, i.e., $(?, r, o, t)$ or $(s, r, ?, t)$, along with a sequence of historical snapshots $\mathcal{G} = \{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_{t-1}\}$ and observation time window m , the proposed model predicts the missing entity (subject or object) based on historical observations $\{\mathcal{G}_{t-m+1}, \dots, \mathcal{G}_{t-1}\}$. Taking the object entity prediction as an example:

$$p(o|s, r, t; \mathcal{G}_{t-m+1:t-1}) = p(o|s, r, t; \mathcal{E}_{t-m+1:t-1}, \mathcal{R}_{t-m+1:t-1}, \mathcal{M}_e, \mathcal{F}_{t-m+1:t-1}) \quad (1)$$

Continuing the example in Fig. 1, given a sequence of historical snapshots $\mathcal{G} = \{\mathcal{G}_a, \mathcal{G}_b, \mathcal{G}_c, \mathcal{G}_d, \mathcal{G}_e\}$, observation time window $m = 4$, and corresponding facts $\{(The\ United\ States, Consult, George\ Bush\ Jr., 2014-03), (The\ United\ States, Consult, George\ Bush\ Sr., 2014-03), (Barack\ Obama, Visit, Europe\ Union, 2014-05), (George\ Bush\ Sr., Visit, Europe\ Union, 2014-05), (George\ Bush\ Jr., Visit, Europe\ Union, 2014-05), (Barack\ Obama, Consult, India, 2014-07)\}$, the proposed model predicts the missing element for the corrupted quadruple $(The\ United\ States, Consult, ?, 2014-11)$, which may occur at a future timestamp.

4. Methodology

This section first presents the overall architecture of MM-TKGF, followed by a comprehensive description of its modules and training objectives. To be specific, the model comprises two modules, i.e., AMFE and TGMF, as presented in Fig. 2. The former AMFE module, which consists of two independent components AGNE and ATE that are detailedly depicted in Fig. 3 and 5 respectively, is developed to identify query-related facts from both structural and temporal views effectively and to learn entity temporal representations. The latter TGMF module, which is thoroughly described in Fig. 6, is designed to integrate multi-modal features with entity temporal representations. Additionally, MM-TKGF takes an MMTKG snapshot sequence as input, and outputs the representations of entities and relations for further forecasting over MMTKGs. Notably, the snapshots from $\mathcal{G}_{t-m+1}$ to $\mathcal{G}_{t-1}$ in Fig. 2 are those to be considered based on the observation time window m .

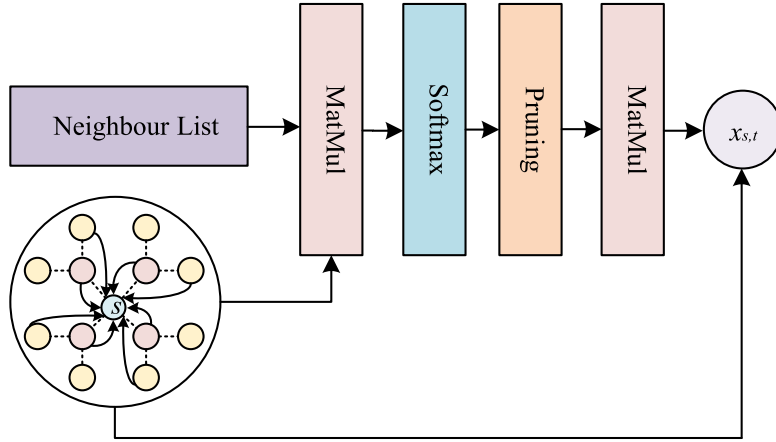


Fig. 3. The architecture of an AGNE unit.

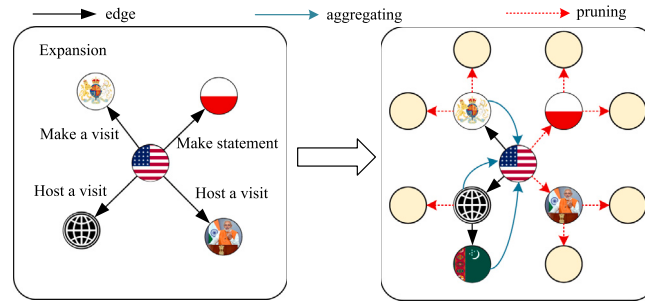


Fig. 4. Illustration of the attentive graph neighbour encoder.

4.1. Attention-based multi-faceted encoder

To address the [Problem 1](#), we design a brand new entity representation module entitled attention-based multi-faceted encoder (AMFE), which captures both entity structural dependencies and temporal evolution simultaneously, and generates temporal representations for entities and relations. To exploit entity information more comprehensively, previous studies ([Chen et al., 2024](#); [Han, Ding, et al., 2021](#); [Li, Sun, & Zhao, 2022](#)) primarily decouple the entity representation into two parts, i.e., structural representation and temporal representation. However, they cannot effectively capture valuable query-relevant information and filter out irrelevant noise from both structural and temporal perspectives. To remedy this, two well-designed independent components are developed, i.e., attentive graph neighbour encoder (AGNE) and attention-based temporal encoder (ATE). Specifically, AGNE filters redundant facts out and extracts the structural information through a self-attention mechanism from the structural perspective. Then, ATE searches query-related facts across historical observations and captures the temporal dynamics of each entity through a self-attention mechanism from the temporal perspective. In what follows, the details of these two components are unfolded.

Attentive Graph Neighbour Encoder. The network aims to filter redundant facts during the training process while identifying valuable structural patterns within the MMTKG snapshot $\mathcal{G}_t$. It outputs the structural representations for entities $\mathcal{X}_{e,t} = \{\vec{x}_{e,t} | e \in \mathcal{E}, t \in \mathcal{T}\}$ and relations $\mathcal{X}_{r,t} = \{\vec{x}_{r,t} | r \in \mathcal{R}, t \in \mathcal{T}\}$. The structure encoder is built through a series of AGNE units, and the detailed design of a unit is illustrated in [Fig. 3](#). Intuitively, the construction of the component draws inspiration from the effectiveness of graph convolutional network in modelling graph structures. Based on this, we use AGNE to generate entity and relation structural representations in each MMTKG snapshot. To better explain the process, we begin to introduce the structural encoder by leveraging an example query $(s, r, ?, t)$, where $?$ denotes a missing element. In detail, the process starts with the query subject entity s and one of its connected relation edges at timestamp t . The input to the structural encoder is initialized entity embedding $\vec{x}_{s,t}^0$ with relation embedding $\vec{x}_{r,t}^0$ in current MMTKG snapshot $\mathcal{G}_t$:

$$\begin{aligned}\vec{x}_{s,t}^0 &= \mathbf{W}_{0,s} \vec{e}_s, \forall t \in \mathcal{T}, \mathcal{T} = \{1, 2, \dots, T\} \\ \vec{x}_{r,t}^0 &= \mathbf{W}_{0,r} \vec{e}_r, \forall t \in \mathcal{T}, \mathcal{T} = \{1, 2, \dots, T\}\end{aligned}\tag{2}$$

where $\vec{e}_s \in \mathbb{R}^d$ denotes as a one-hot vector for entity s , $\vec{e}_r \in \mathbb{R}^d$ represents a one-hot vector for relation r , $\mathbf{W}_{0,s} \in \mathbb{R}^{d \times |\mathcal{E}|}$ is entity embedding matrix, and $\mathbf{W}_{0,r} \in \mathbb{R}^{d \times |\mathcal{R}|}$ is relation embedding matrix.

Next, to measure which neighbours are more important to the query within an MMTKG snapshot, we design a query-dependent attention mechanism. The aggregation process illustrated in Fig. 4 proceeds as follows. Given a query $(s, r, ?, t)$, we create a self-loop quadruple (s, r_0, s, t) and add it into the neighbour list L_s^l , where r_0 represents the self-loop relation. For each object entity $p \in L_s^l$, i adjacent relations $\{j_0^l, j_1^l, \dots, j_i^l\} \in \mathcal{R}_t$ and their corresponding connected entities $\{q_0^l, q_1^l, \dots, q_i^l\}$ are subsequently sampled. These sampled relation–entity pairs are further formed into quadruples, which are presumed to be relevant to the query, and appended to the neighbour list L_s^l ,

$$L_s^l = \{(p, j, q, t) | (e_n, r_n, p, t) \in L_s^{l-1}\} \cup \{(s, r_0, s, t)\} \quad (3)$$

where l represents the number of expansion layers, $l=1$ corresponds to aggregating single-hop neighbours for the query s , $\{(e_n, r_n, p, t)\}$ denotes the set of quadruples stored in neighbour list L_s^{l-1} and their object entities are further used as subject entities for the next expansion. To dynamically weight sampled neighbours in L_s^l , the query-dependent attention scores are calculated via the equation below, enabling adaptive aggregation based on the relevance of neighbours to the query.

$$e_{p,j,q}^l = \mathbf{W}_{sub}^l (\tilde{x}_{p,t}^{l-1} \parallel \tilde{x}_{s,t}^{l-1} \parallel \tilde{x}_{r,t}^{l-1}) \cdot \mathbf{W}_{obj}^l (\tilde{x}_{q,t}^{l-1} \parallel \tilde{x}_{s,t}^{l-1} \parallel \tilde{x}_{r,t}^{l-1} \parallel \tilde{x}_{j,t}^{l-1}) \quad (4)$$

where $e_{p,j,q}^l$ is the attention score for expansion quadruple (p, j, q, t) corresponding to the query $(s, r, ?, t)$, $\parallel$ represents the concatenation operation, $\mathbf{W}_{sub}^l$ and $\mathbf{W}_{obj}^l$ denote weight matrices that capture the structural dependencies between query and expansion quadruple. Subsequently, the normalized attention score $w_{p,j,q}$ for each neighbour is computed via the soft-max function as follows:

$$w_{p,j,q} = \frac{\exp(e_{p,j,q}^l)}{\sum_{(p,j,q,t) \in L_s^l} \exp(e_{p,j,q}^l)} \quad (5)$$

Following the calculation of attention scores, we analyse which neighbours provide valuable structural information. With the expansion proceeding, the neighbour list may store entities with low attention scores, thereby increasing the computational cost of AGNE layer. To mitigate this, we prune redundant neighbours (i.e., those that contribute less structural information) by retaining k neighbours with the highest scores. Specifically, neighbours are sorted by descending attention scores, and a cut-off number k is used to determine how many neighbours should be retained to provide valuable structural information for the query.

$$L_s^l = \{(p, j, q, t) \in L_s^l \setminus \{(s, r_0, s, t)\} \mid \mathbb{1}(\text{rank}(w_{p,j,q} | L_s^l) < k)\} \cup \{(s, r_0, s, t)\} \quad (6)$$

where $\mathbb{1}$ represents the binary indicator function, and the function $\text{rank}(\cdot)$ denotes the position of a quadruple in the neighbour list L_s^l that is sorted based on the normalized attention score $w_{p,j,q}$ in descending order. Notably, in addition to single hop neighbours, structural information also incorporates further neighbours at multi-hops (e.g. 2-hops or more). In order to obtain further structural dependencies, AGNE aggregates l -hops neighbour entities by iteratively applying the process l times.

By applying the aforementioned attention mechanism with adaptive edge pruning, AGNE extracts the most significant structural dependencies of entity s from its sampled neighbours. Subsequently, the neighbour information is aggregated to iteratively update the structural representation of s via the given equation:

$$\tilde{x}_{s,t}^l = \sigma \left(\sum_{j_0, \dots, j_i \in \mathcal{R}_t} \sum_{(p,j,q,t) \in L_s^l \setminus \{(s,r_0,s,t)\}} \frac{1}{|j_i|} w_{p,j,q} \mathbf{W}_{att,i}^l \tilde{x}_{p,t}^{l-1} + \mathbf{W}_s^l \tilde{x}_{s,t}^{l-1} \right) \quad (7)$$

where $\tilde{x}_{p,t}^{l-1}$ is the structural representation of neighbour p for layer $l-1$, $\sigma(\cdot)$ represents the LeakyReLU activation function, $\mathbf{W}_{att,i}^l$ denotes the relation-specific transformation matrix for layer l , $\mathbf{W}_s^l$ represents the transformation matrix for layer l , and matrices $\mathbf{W}_{att,i}^l, \mathbf{W}_s^l$ are shared at different timestamps. The final structural representation $\tilde{x}_{s,t}^l$ of entity s at timestamp t is obtained by passing $\tilde{x}_{s,t}^l$ through a dense layer.

Algorithm 1 provides an AGNE unit detailedly through pseudo-code, which captures the structural dependencies within an MMTKG snapshot. To characterize entities more thoroughly, it is essential to model their evolution across multiple MMTKG snapshots. Therefore, the structural representations are then fed into the ATE module to extract the temporal dynamics subsequently.

Attention-based Temporal Encoder. This encoder aims to model the entity evolution by capturing their valuable temporal dynamics across MMTKG snapshots, and outputs the set of entity temporal representations. Fig. 5 presents the details of an ATE unit. In particular, in this study, we assume that the relations are static, meaning that their semantics remain unchanged over time. Therefore, the evolution of relational semantics is not considered (Jin et al., 2020; Wang et al., 2022).

Intuitively, the design of ATE derives inspiration from the proven success of attention mechanisms in event sequence modelling (Han et al., 2020). In detail, this mechanism enables the model to emphasize query-related events within longer temporal sequence. The learning units are designed sequentially, with each unit capturing the temporal representations of entities at a specific timestamp. ATE outputs an entity temporal representation set $\mathcal{H}_{e,t} = \{h_{e,t} | e \in \mathcal{E}, t \in \mathcal{T}\}$, capturing the entity evolution from initial to the current timestamp t . Continuing the example query $(s, r, ?, t)$ in AGNE component, we feed the structural representation $\tilde{x}_{s,t}$ of s at timestamp t into ATE to start the process. Specially, if the query occurs at the timestamp $t = 1$ (i.e., the first timestamp), the structural representation $\tilde{x}_{s,t}$ serves as the initial temporal representation $\tilde{h}_{s,t}$. We first search for a set of its past temporal representations at the most recent m timestamps, i.e., from timestamp $t - m + 1$ to $t - 1$, to generate the temporal representation $\tilde{h}_{s,t}$ for the current timestamp. The reason for only considering the limited set of recent m snapshots is that the characteristics of the entities are more closely related to recent observations (Han, Ding, et al., 2021). As assumed in AGNE, we also hypothesize that not all past temporal representations are equally relevant to the query, so it is necessary to highlight valuable temporal facts from

Algorithm 1 AGNE

Input: An MMTKG snapshot $\mathcal{G}_t$ at timestamp t and the number of expansion layers l

Output: The set of structural representations $\mathcal{X}_{e,t} = \{\vec{x}_{e,t} | e \in \mathcal{E}_t, t \in \mathcal{T}\}$

for $(s, r, ?, t) \in \mathcal{G}_t$ **do**

 Initializing a neighbour list for entity s : $L_s^0 = \{(s, r_0, s, t)\}$;

for $c = 1, l$ **do**

 For each entity p in list L_s^c , search the neighbour entities $p_i \in \mathcal{G}_t$, and add them into list L_s^c ;

 For each c -hop(s) neighbour (p_i, j_i, q_i, t) , the query-dependent attention score is computed via Eq. (5);

 Attention scores are sorted by descending order;

 Retain k neighbours (p_i, j_i, q_i, t) with the highest scores and update list L_s^c via Eq. (6);

 Update the structural representation $\vec{x}_{s,t}^c$ via Eq. (7);

$c \leftarrow c + 1$;

end for

 Update the structural representation $\vec{x}_{s,t}$ of entity s ;

 Update $\vec{x}_{s,t}$ in set $\mathcal{X}_{e,t} = \{\vec{x}_{1,t}, \vec{x}_{2,t}, \dots, \vec{x}_{s,t}, \dots, \vec{x}_{n,t}\}$;

end for

return $\mathcal{X}_{e,t} = \{\vec{x}_{e,t} | e \in \mathcal{E}_t, t \in \mathcal{T}\}$;

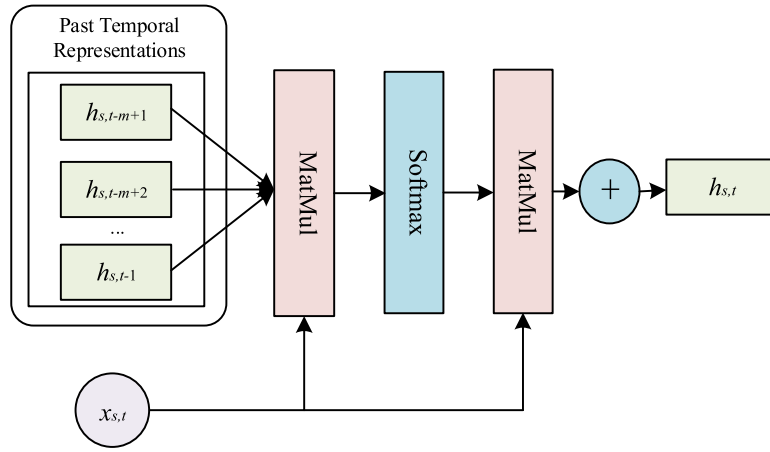


Fig. 5. The details of an ATE unit.

historical MMTKG snapshots. Thus, we assign varying importance to each snapshot at timestamp $\tau \in [t - m + 1, t - 1]$ by calculating an attention score as follows:

$$\begin{aligned}
 u_{s,t,t-\tau} &= \frac{(\vec{x}_{s,t} \mathbf{W}_q)(\vec{h}_{s,t-\tau} \mathbf{W}_k)^\top}{\sqrt{d}} \\
 \gamma_{s,t,t-\tau} &= \begin{cases} 1 & \text{if entity } s \text{ is present at timestamp } t - \tau \\ 0 & \text{otherwise} \end{cases} \\
 \beta_{s,t,t-\tau} &= \frac{\exp(u_{s,t,t-\tau} \gamma_{s,t,t-\tau})}{\sum_{i=0}^m \exp(u_{s,t,t-i} \gamma_{s,t,t-i})} \\
 \vec{h}_{s,t} &= \sum_{\tau=0}^m \beta_{s,t,t-\tau} (\vec{h}_{s,t-\tau} \mathbf{W}_v) + \vec{x}_{s,t}
 \end{aligned} \tag{8}$$

where $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$ are linear projection matrices, β denotes the self-attention score, m denotes the observation time window, and γ is a binary indicator representing whether an entity is present or absent at timestamp t . To be specific, γ is assigned to 1 if the entity exists in any quadruple at timestamp t , and 0 otherwise.

Upon completing the above steps, ATE generates the entity temporal representations at each timestamp. Algorithm 2 presents an ATE unit intuitively through pseudo-code, which captures the temporal evolution of entities across different MMTKG snapshots thoroughly. Although capturing the information of entities from both local structural and global temporal views can accurately predict most temporal facts, as discussed in Section 1, we recognize that relying solely on graph structure cannot distinguish similar entities effectively. Therefore, a viable solution is to leverage multi-modal information to enhance entity representations.

Algorithm 2 ATE

Input: An MMTKG historical snapshot sequence $\mathcal{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_{t-1}\}$, and observation time window m

Output: The set of temporal representations $\mathcal{H}_{e,t} = \{\bar{h}_{e,t} | e \in \mathcal{E}_t, t \in \mathcal{T}\}$

for $(s, r, ?, t) \in \mathcal{G}_t$ **do**

for $i = 0, m$ **do**

if timestamp = 1, set the structural representation $\bar{x}_{s,t-m+1}$ as initial temporal representation $\bar{h}_{s,t-m+1}$;

 Compute self-attention score $\beta_{t-m+1,t-i}$;

$i \leftarrow i + 1$;

end for

 Update temporal representation $\bar{h}_{s,t}$ via Eq. (8);

 Update $\bar{h}_{s,t}$ in set $\mathcal{H}_{e,t} = \{\bar{h}_{1,t}, \bar{h}_{2,t}, \dots, \bar{h}_{s,t}, \dots, \bar{h}_{n,t}\}$;

end for

return $\mathcal{H}_{e,t} = \{\bar{h}_{e,t} | e \in \mathcal{E}_t, t \in \mathcal{T}\}$;

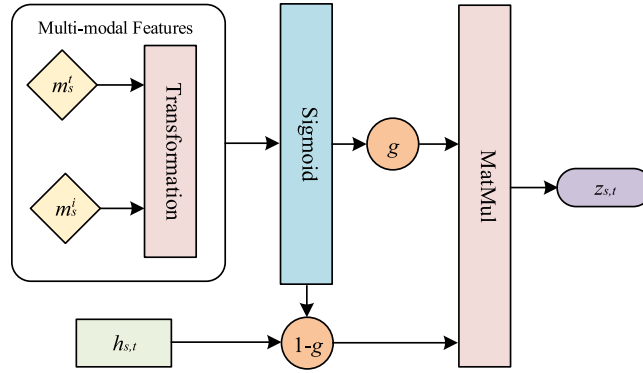


Fig. 6. The details of a unit of TGMF.

Consequently, these temporal representations are subsequently fed into the following temporal-aware gated multi-modal fusion module.

4.2. Temporal-aware gated multi-modal fusion module

To tackle the [Problem II](#), we design a new fusion strategy called temporal-aware gated multi-modal fusion module (TGMF). The design of this module is motivated by the previous MMKG reasoning studies ([Liu et al., 2019](#); [Wei et al., 2023](#); [Zheng et al., 2023](#)), which collectively focus on fusing static representations with multi-modal features. However, their fusion strategy is inadequate for dynamic temporal representations as it may risk compromising temporal representations due to their early fusion strategy. To address this, TGMF is specifically designed to integrate temporal representations with multi-modal features, automatically balancing their contributions via a gating mechanism to enable fine-grained fusion. In detail, TGMF is composed of three parts: multi-modal data preparation, multi-modal feature extraction, and a series of gated fusion units. By arranging the fusion units sequentially, each TGMF unit integrates the temporal representations of entities with their corresponding multi-modal features at each timestamp. TGMF outputs an entity unified representations set $\mathcal{Z}_{e,t} = \{\bar{z}_{e,t} | e \in \mathcal{E}_t, t \in \mathcal{T}\}$. [Fig. 6](#) presents a TGMF unit, with further details explained below.

Multi-modal Data Preparation. Two types of data are employed as multi-modal auxiliary information, which are detailed below. (1) Textual information: this category encompasses the fundamental attributes and numerical data of entities, which reveals the core characteristics of entities. Previous MMKG studies (e.g., [Liu et al. \(2019\)](#) and [Xie et al. \(2016\)](#)) have demonstrated that textual descriptions and text-based attributes can significantly enhance entity representation. Furthermore, numerical literals are also important for the construction of MMKGs ([Liu et al., 2019](#)). To extract the textual information associated with each entity, we align entities in ICEWS with those from Wikidata (i.e., align entities by string matching and manual annotation). Then, for each entity, we explore its attributes extracted from Wikidata web pages, and identify several key attributes including *instance of*, *coordinate location*, *country*, and numerical values. (2) Image information: Images are the most common type of data on the Internet, containing vivid descriptions of entities, and are widely used in the construction of MMKGs ([Liu et al., 2019](#); [Wang et al., 2020](#)). To acquire images related to each entity, the data on Wikidata is utilized as the multi-modal data source, and ICEWS entities are aligned with them. Next, for each entity, the abnormal, duplicate, and irrelevant images are filtered out following the idea of [Wang et al. \(2020\)](#) and [Liu et al. \(2019\)](#).

Multi-modal Feature Extraction. To fully leverage the multi-modal information, the following pre-trained models are used to extract entity representations on different modalities. (1) Textual feature vectors $\mathcal{M}_e^t = \{\tilde{m}_e^t | e \in \mathcal{E}\}$ for all entities are generated by using the pre-trained model Doc2vec (Le & Mikolov, 2014), where the textual feature m_e^t of entity e is expressed into a d_t dimensions vector. (2) Image feature vectors $\mathcal{M}_e^i = \{\tilde{m}_e^i | e \in \mathcal{E}\}$ for all entities are obtained by utilizing the pre-trained model ResNet (He et al., 2016), with its last layer removed, and the image feature m_e^i of entity e is represented as a d_i dimensions vector. Next, for each entity $e \in \mathcal{E}$, we concatenate two groups of multi-modal features $\tilde{m}_e^i$ and $\tilde{m}_e^t$ to form the multi-modal auxiliary feature $\tilde{m}_e \in \mathcal{M}_e$:

$$\begin{aligned} \tilde{m}_e &= \sigma(\mathbf{W}_m(\mathbf{W}_i \tilde{m}_e^i \parallel \mathbf{W}_t \tilde{m}_e^t) + \tilde{b}_m) \\ \mathcal{M}_e^t, \mathcal{M}_e^i &\subset \mathcal{M}_e = \{\tilde{m}_{e_1}, \tilde{m}_{e_2}, \dots, \tilde{m}_{e_n}\} \end{aligned} \quad (9)$$

where $\mathbf{W}_m, \mathbf{W}_i, \mathbf{W}_t$ are embedding matrices, $\tilde{m}_e$ is the multi-modal representation of entity e , $\sigma(\cdot)$ denotes the *sigmoid* activation function, and $\parallel$ represents the concatenation operation. Due to the higher dimensionality of multi-modal features compared to temporal representations, we concatenate them first, then project the multi-modal features through a linear transformation layer to transform them into d -dimensional vectors.

Multi-modal Feature Fusion. The fusion unit aims to exploit multi-modal features from various modalities to enhance entity temporal representations, and outputs the entity unified representations set $\mathcal{Z}_{e,t} = \{\tilde{z}_{e,t} | e \in \mathcal{E}, t \in \mathcal{T}\}$ at each timestamp. In this process, the entity temporal representations $\mathcal{H}_{e,t}$ are fused with multi-modal feature vectors $\mathcal{M}_e$ at the middle layer of MM-TKGF (i.e., intermediate fusion), and the reason for not performing the direct fusion of $\mathcal{M}_e$ with structural representations $\mathcal{X}_{e,t} = \{\tilde{x}_{e,t} | e \in \mathcal{E}, t \in \mathcal{T}\}$ (i.e., early fusion) is that such operation might compromise the quality of temporal representations $\mathcal{H}_{e,t}$. This is because the structural representations at different timestamps are repeatedly integrated with static multi-modal features, resulting in a high vector similarity between the fused representations. In addition, direct fusion (i.e., concatenation or addition operation) may also degrade the quality of entity representations, as it cannot filter out inter-modal noise (Zheng et al., 2023). To solve the problem, a gating unit is utilized here to filter the noise out and achieve a fine-grained fusion. For instance, given an entity e , we fuse the temporal representation $\tilde{h}_{e,t}$ with its corresponding multi-modal feature $\tilde{m}_e$ as follows:

$$\begin{aligned} g_t &= \sigma(\mathbf{W}_g \tilde{m}_e + \tilde{b}_g) \\ \tilde{z}_{e,t} &= (1 - g_t) \tilde{h}_{e,t} + g_t \tilde{m}_e \end{aligned} \quad (10)$$

where $\mathbf{W}_g$ represents the gating matrix, $\tilde{z}_{e,t}$ denotes the unified representation of entity e at timestamp t , $\sigma(\cdot)$ is the *sigmoid* activation function, and $\tilde{b}_g$ is a scalar bias. g_t and $1 - g_t$ are gating parameters that are used to balance the ratio of temporal representation and multi-modal features. Based on the gating mechanism, the model automatically adjusts the proportion of different source features and eliminates the inter-modal noise, then extracts the valuable information through a fine-grained fusion way. After these steps, the unified representations $\mathcal{Z}_{e,t}$ are effectively enhanced by a variety of multi-modal features.

4.3. Training objective

The entity prediction aims to minimize the discrepancy between predicted value and ground truth by leveraging both graph structure and multi-modal features. Thus, the training objective includes the following two parts. For the graph part, we adopt the ConvTransE decoder as the scoring function (Shang et al., 2019), which has been widely utilized in recent TKG studies (Chen et al., 2024; Li, Hou, et al., 2022). Similarly, for the multi-modal part, we also utilize this decoder. Given a query $(s, r, ?, t)$, the object entity prediction is generally formulated as a multi-class classification problem, where each type of entity represents a class. Specifically, we parameterize this task as follows:

$$\begin{aligned} p(o|s, r, t; \mathcal{G}_{t-m:t-1}) &\propto \phi(s, r, o, t) \\ \phi(s, r, o, t) &= \alpha \cdot \sigma(\tilde{h}_{o,t} \text{ConvTransE}(\tilde{h}_{s,t}, \tilde{x}_{r,t})) + (1 - \alpha) \cdot \sigma(\tilde{z}_{o,t} \text{ConvTransE}(\tilde{z}_{s,t}, \tilde{x}_{r,t})) \\ o &= \arg \max_{o \in \mathcal{G}} p(o|s, r, t; \mathcal{G}_{t-m:t-1}) \end{aligned} \quad (11)$$

where $\text{ConvTransE}(\cdot)$ indicates ConvTransE decoder, $\sigma(\cdot)$ represents the *sigmoid* activation function, ϕ denotes the hybrid form of ConvTransE decoder, and α is a learnable parameter to modulate the balance between two terms within the hybrid decoder ϕ . By symmetry, the inference process can also be extended to predict subject entities.

$$\begin{aligned} p(s|o, r, t; \mathcal{G}_{t-m:t-1}) &\propto \phi(s, r, o, t) \\ s &= \arg \max_{s \in \mathcal{G}} p(s|o, r, t; \mathcal{G}_{t-m:t-1}) \end{aligned} \quad (12)$$

The loss function for object entity prediction, corresponding to the training objective, is formally defined below:

$$\mathcal{L}_1 = - \sum_t \sum_{(s,r) \in \mathcal{G}} \mathbb{1}_{(s,r,o,t) \in \mathcal{G}} \log p(o|s, r, t; \mathcal{G}_{t-m:t-1}) \quad (13)$$

where $p(o|s, r, t; \mathcal{G}_{t-m:t-1})$ denotes the probability over all candidate entities, $\mathbb{1}$ represents the binary indicator. Similar to object entity prediction, we symmetrically define the loss function of subject entity prediction as below:

$$\mathcal{L}_2 = - \sum_t \sum_{(o,r) \in \mathcal{G}} \mathbb{1}_{(s,r,o,t) \in \mathcal{G}} \log p(s|o, r, t; \mathcal{G}_{t-m:t-1}) \quad (14)$$

The training loss of model is computed as the total of losses for two prediction tasks, i.e., $\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2$.

Table 2
Statistics of datasets.

Dataset	$ \mathcal{E} $	$ \mathcal{R} $	$ \mathcal{F} $	$ \mathcal{T}_g $	$ \mathcal{T}_s $	ρ	$ \mathcal{M}_e^t $	$ \mathcal{M}_e^v $	#Training	#Validation	#Test
MMICEWS'14	10 261	235	264 958	24 h.	365 days	725.9	10 261	7294	211 268	27 689	26 001
MMICEWS'18	7665	229	297 663	24 h.	365 days	814.8	7665	5764	238 636	29 360	29 397
MMICEWS'05-15	14 432	246	603 090	24 h.	4017 days	150.1	14 432	10 099	482 314	61 520	59 256

5. Experiments

To fully investigate the performance of MM-TKGF, three novel MMTKGs are constructed, on which comprehensive experiments are conducted. Further details are introduced as follows.

5.1. Research questions

RQ1: How does MM-TKGF perform compared to the existing KG, MMKG, and TKG reasoning methods?

RQ2: Which components of MM-TKGF contribute to its effectiveness?

RQ3: To what extent do hyper-parameters impact the performance of MM-TKGF?

RQ4: Are there any intuitive cases that qualitatively illustrate the effectiveness of MM-TKGF?

RQ5: What is the computational efficiency of MM-TKGF?

5.2. Experimental setup

Datasets. Having recognized the limitations of current TKG datasets that only contain structural data, we construct three new multi-modal temporal knowledge graph datasets, namely MMICEWS'14, MMICEWS'18, and MMICEWS'05-15, to comprehensively evaluate the impact of multi-modal auxiliary information on MMTKG forecasting. These constructed MMTKGs,³ which are subsets of ICEWS and contain the temporal facts in 2014, 2018, and 2005–2015, leverage the graph structure of existing TKG datasets (i.e., ICEWS'14, ICEWS'05-15, and ICEWS'18) as a foundation, and enrich them with multi-modal data from Wikidata⁴ (Han, Chen, et al., 2021; Li, Guan, et al., 2022). Detailedly, during the construction of MMTKGs, we align the multi-modal data (i.e., images and text) with corresponding entities with the following methods. For image-based information, we focus on images such as *photographs*, *logotypes*, *maps*, *emblems*, and *signatures*, rescaling them all to a uniform size of 224×224 pixels. For textual-based information, we employ entity attributes such as *instance of*, *coordinate location*, *country*, and numerical values. Table 2 summarizes the statistics of these datasets, where $|\mathcal{E}|$ represents entity counts, $|\mathcal{R}|$ denotes relation counts, $|\mathcal{F}|$ is the fact counts, $|\mathcal{T}|_g$ represents time granularity, $|\mathcal{T}|_s$ denotes time-span, ρ is the number of facts per time granularity, $|\mathcal{M}_e^t|$ and $|\mathcal{M}_e^v|$ are the number of entities containing their respective textual and vision information, and #Training, #Validation, #Test represent quadruple counts in the training, validation, test sets respectively.

Baselines. As our proposed model MM-TKGF is the first attempt to solve MMTKG forecasting, a comprehensive comparison necessitates including three categories of baselines, as listed follows: (1) Uni-modal static methods. These approaches, designed for static KGs, are not directly applicable to MMTKGs, as they cannot leverage temporal information and multi-modal data. To make them compatible with our datasets, we discard the multi-modal data and timestamp, and build a static KG that encompasses all facts, aligning with the setup used in existing TKG studies (Jin et al., 2020; Zhu et al., 2021). To be specific, these efforts involve RotatE (Sun et al., 2019), DistMult (Goel et al., 2020), ConvE (Dettmers et al., 2018), and AnyBURL (Meilicke et al., 2019). (2) Multi-modal static methods. We also benchmark our model against existing MMKG reasoning models, which are intended for MMKGs and not directly applicable to the MMTKG scenario. To adapt these methods to MMTKG datasets, we construct a static MMKG by preserving all multi-modal triplets while eliminating temporal information. These studies include MKGFormer (Chen et al., 2022), TransAE (Mousselly-Sergieh et al., 2018), IKRL (Xie et al., 2017), Native (Zhang, Chen, et al., 2024), SNAG (Chen et al., 2025), and SGMP (Liang et al., 2024). (3) Uni-modal temporal methods. MM-TKGF is further compared with existing uni-modal TKG reasoning methods, including CyGNet (Zhu et al., 2021), TANGO (Han, Ding, et al., 2021), Know-Evolve (Trivedi et al., 2017), EvoKG (Park et al., 2022), CEN (Li, Guan, et al., 2022), STDN (Li et al., 2024), CENET (Xu et al., 2023), TiRGN (Li, Sun, & Zhao, 2022), Re-Temp (Wang, Han, & Poon, 2023), PLEASING (Zhang, Sun, et al., 2024), and LogCL (Chen et al., 2024) which are developed for TKGs and cannot directly incorporate multi-modal features. To adapt these approaches to MMTKGs, we remove multi-modal data, and build a uni-modal TKG by preserving all quadruples. (4) The model TKGF-NTP proposed in our conference version (Han et al., 2023). To further verify that multi-modal data can improve the performance of uni-modal TKG reasoning model, we also compare MM-TKGF to TKGF-NTP. Notably, to ensure fairness, the experimental results for all baselines are produced using their official open-source codes with their inherent parameter settings and configurations.

Model Configurations. In the training process, the model parameters are initialized with Xavier initialization method and optimized using AMSGrad optimizer. The initial learning rates are set to 0.0001 for MMICEWS'14, 0.0001 for MMICEWS'18, and

³ <https://github.com/ikasimb/MMTKGF/>

⁴ <https://wikidata.org/>

Table 3
Results of link prediction on all datasets.

	Models	MMICEWS'14				MMICEWS'18				MMICEWS'05-15			
		MRR	Hits@1	Hits@3	Hits@5	MRR	Hits@1	Hits@3	Hits@5	MRR	Hits@1	Hits@3	Hits@5
Static	RotatE	18.74	10.43	21.04	22.82	13.64	6.05	17.85	21.43	20.64	11.27	23.62	26.52
	DistMult	17.03	7.36	19.13	20.58	16.24	5.46	20.42	23.85	18.45	8.74	18.77	20.38
	ConvE	18.83	12.89	21.75	24.42	19.43	10.46	24.47	29.46	24.58	15.50	26.85	28.45
	AnyBURL	23.96	12.72	25.37	28.63	24.67	15.43	31.46	34.63	30.80	21.56	31.74	34.51
Multi-modal	TransAE	35.74	28.80	36.95	40.23	24.32	16.06	28.68	31.85	35.34	28.07	37.51	41.50
	IKRL	30.47	23.82	34.52	38.27	28.16	18.29	31.63	36.56	32.92	27.77	35.83	39.05
	MKGFormer	39.42	28.53	44.72	49.24	28.74	19.52	31.62	37.93	32.73	24.72	36.62	40.24
	NativE	37.61	26.83	39.52	41.84	29.59	18.44	30.96	35.14	34.18	25.47	36.88	40.76
	SNAG	32.81	23.93	35.67	37.08	26.93	17.26	30.77	34.41	32.22	23.31	35.04	38.62
Temporal	SGMPT	36.87	26.13	40.37	45.61	28.37	18.45	31.21	35.83	32.94	24.57	36.92	40.33
	Know-Evolve	27.29	17.15	30.67	34.83	18.64	10.45	21.83	25.96	21.64	13.82	23.71	29.59
	CyGNet	34.66	23.91	38.24	43.72	26.54	16.26	30.36	36.41	30.72	22.13	35.77	39.93
	TANGO-TuckER	32.70	23.81	35.33	40.12	24.63	17.79	29.84	34.70	27.29	18.87	31.68	36.94
	TANGO-DistMult	33.12	24.07	37.28	41.85	27.84	16.95	31.28	36.27	28.88	18.42	32.03	39.22
	EvoKG	29.76	22.51	34.52	39.48	26.32	17.42	32.35	37.61	27.74	16.85	31.43	36.76
	CEN	38.74	30.56	45.60	49.37	30.29	22.92	35.83	40.96	34.32	24.86	37.28	41.95
	STDN	35.27	28.14	40.93	46.23	28.49	22.06	34.07	38.46	35.89	28.18	41.57	46.34
	CENET	39.06	31.52	44.28	49.29	31.92	24.18	37.89	41.87	35.92	28.37	40.96	45.26
	TIRGN	35.62	24.89	39.32	45.83	27.40	17.21	32.69	37.18	32.03	22.84	36.40	42.37
	LogCL	41.72	33.29	45.85	50.23	33.29	23.26	38.23	41.29	37.25	27.92	42.88	47.84
	Re-Temp	30.52	21.42	34.17	38.83	24.84	13.68	29.12	33.83	31.76	21.12	35.06	38.18
	PLEASING	36.31	27.15	40.54	44.96	29.76	19.43	33.93	38.06	33.52	24.17	37.32	42.13
	TKGF-NTP	35.74	27.21	40.62	46.01	29.62	22.63	33.06	38.42	32.73	26.23	35.27	41.87
	MM-TKGF	42.34	34.01	46.23	51.50	35.74	26.58	39.72	43.74	39.85	31.57	44.72	49.24

* The best results are highlighted in bold.

** The results are presented as percentages.

0.001 for MMICEWS'05-15. For negative sampling strategy, we follow the partial completeness assumption (Galárraga et al., 2013). Specifically, for each positive sample, we generate a negative sample by corrupting either the subject or object entity through random replacement with another entity, and we set the negative sampling ratio to 1. The textual features of entities are pre-trained by Doc2vec and represented as 200-dimensional vectors. Similarly, image features are pre-trained by ResNet-152, with its last soft-max layer removed, and extracted as 2048-dimensional vectors. We set the batch size to 64, and the dimensions of structural embeddings to 256. The experiments are performed on a Linux server with Intel Xeon@E5-2680v4 CPUs and NVIDIA 1080Ti GPUs. The model implementation leverages Python 3.6.10, with PyTorch and DGL for neural network parts.

Evaluation Setting and Metrics. To ensure a fair evaluation, we adhere to the evaluation setting from existing TKG studies (Han, Chen, et al., 2021; Xu et al., 2023) and our conference version (Han et al., 2023). Specifically, the dataset is partitioned with an 8: 1: 1 ratio for training, validation, and test sets respectively, preserving the chronological order such that: $t_{\text{training}} < t_{\text{validation}} < t_{\text{test}}$. To maintain the coherence of timestamps, each of them includes at least one quadruple. In addition, to evaluate both subject or object entity prediction tasks, we also augment the validation and test sets by adding reciprocal quadruples (Jin et al., 2020). Notably, hyper-parameters are tuned via grid search on the validation set, with optimal values selected based on the best performance.

With respect to evaluation metrics, we use a filtered version of Mean Reciprocal Ranks (MRR) and Hits rate(s), i.e., Hits@ k ($k = 1, 3, 5$), which are commonly employed to measure the performance of the TKG reasoning task. To be specific, MRR calculates the average inverse rank of the correct entity for all queries in the test set, while Hits@ k measures the proportion of queries where the correct answers appear in the top- k ranked predictions, following the setting in prior work (Bordes et al., 2013). During the evaluation, we exclude all quadruples present in the training set, and only utilize facts from the validation and test sets as queries. Furthermore, all metrics in this paper are presented as percentages.

5.3. Performance comparison on MMTKGs (RQ1)

Main Results. The experimental results of MM-TKGF and all baselines, which are reported with standard evaluation metrics (average over 5 runs), are presented in Table 3. As observed, MM-TKGF consistently outperforms existing baselines across all three MMTKG datasets. The most substantial improvements are achieved on MMICEWS'18, where MM-TKGF attains gains of +7.36% in MRR, +9.93% in Hits@1, +3.90% in Hits@3, and +4.47% in Hits@5. On MMICEWS'05-15, the model demonstrates a similarly strong performance, with MRR, and Hits@1, 3, 5 increasing by +6.98%, +11.28%, +5.29%, and +2.86%, respectively. While still surpassing baselines, the gains on MMICEWS'14 are comparatively modest, where MM-TKGF improves MRR by +1.49%, Hits@1 by +2.16%, Hits@3 by +0.83%, and Hits@5 by +2.53%. These experimental results collectively highlight the superior performance of our proposed model MM-TKGF in MMTKG datasets.

Discussions. To analyse the superiority of MM-TKGF thoroughly, we benchmark its performance against four categories of the strongest baselines in Table 3: static methods (e.g., AnyBURL), multi-modal static methods (e.g., MKGFormer), uni-modal temporal

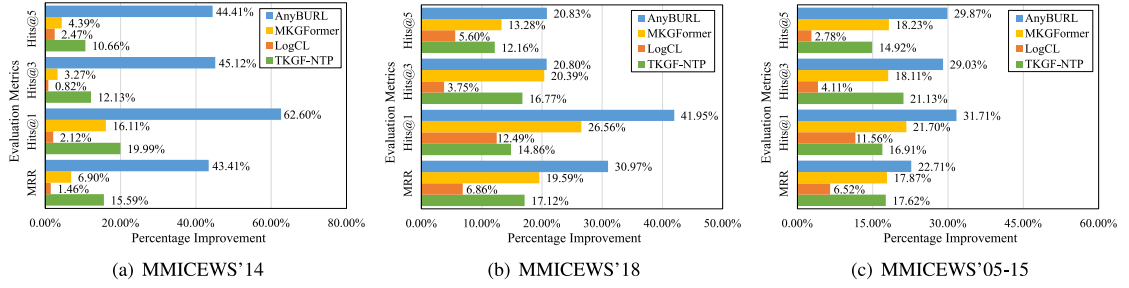


Fig. 7. Comparative analysis of four categories of baselines.

Table 4

Performance comparison of uni-modal temporal reasoning methods with their multi-modal augmentation variants.

	MMICEWS'14				MMICEWS'18				MMICEWS'05-15			
	MRR	Hits@1	Hits@3	Hits@5	MRR	Hits@1	Hits@3	Hits@5	MRR	Hits@1	Hits@3	Hits@5
Know-Evolve	27.29	17.15	30.67	34.83	18.64	10.45	21.83	25.96	21.64	13.82	23.71	29.59
+MM	23.80	12.78	27.57	33.01	15.52	9.44	19.46	23.62	21.84	13.70	24.34	28.63
CyGNet	34.66	23.91	38.24	43.72	26.54	16.26	30.36	36.41	30.72	22.13	35.77	39.93
+MM	37.76	28.98	39.24	44.43	29.32	18.75	29.53	32.52	34.26	27.64	36.29	43.87
TANGO-TuckER	32.70	23.81	35.33	40.12	24.63	17.79	29.84	34.70	27.29	18.87	31.68	36.94
+MM	36.81	27.72	39.17	44.18	28.61	17.67	32.27	36.49	29.68	24.62	35.64	43.56
TANGO-DistMult	33.12	24.07	37.28	41.85	27.84	16.95	31.28	36.27	28.88	18.42	32.03	39.22
+MM	34.60	26.95	38.03	40.46	29.23	18.72	31.73	37.32	30.11	25.03	35.33	44.52
EvoKG	29.76	22.51	34.52	39.48	26.32	17.42	32.35	37.61	27.74	16.85	31.43	36.76
+MM	31.84	25.00	36.01	40.32	28.66	18.78	32.36	38.48	30.46	19.58	32.52	37.54
CEN	38.74	30.56	45.60	49.37	30.29	22.92	35.83	40.96	34.32	24.86	37.28	41.95
+MM	41.52	32.63	46.04	51.47	31.62	23.62	36.52	42.72	35.37	26.42	38.52	43.62
STDN	35.27	28.14	40.93	46.23	28.49	22.06	34.07	38.46	35.89	28.18	41.57	46.34
+MM	34.93	26.04	40.23	45.82	28.72	21.87	32.75	37.75	37.66	30.53	42.72	46.89
CENET	39.06	31.52	44.28	49.29	31.92	24.18	37.89	41.87	35.92	28.37	40.96	45.26
+MM	39.71	32.58	44.82	50.01	33.62	24.61	39.71	45.37	36.50	29.86	40.24	45.67
TiRGN	35.62	24.89	39.32	45.83	27.40	17.21	32.69	37.18	32.03	22.84	36.40	42.37
+MM	36.23	26.08	41.77	47.06	28.63	19.74	35.16	39.22	33.15	23.16	37.28	44.14
LogCL	41.72	33.29	45.85	50.23	33.29	23.26	38.23	41.29	37.25	27.92	42.88	47.84
+MM	41.86	33.75	46.38	51.37	34.25	23.81	38.72	42.69	38.05	28.34	43.20	48.39
Re-Temp	30.52	21.42	34.17	38.83	24.84	13.68	29.12	33.83	31.76	21.12	35.06	38.18
+MM	27.86	16.30	30.04	33.98	24.61	13.74	28.66	33.16	32.94	22.77	37.15	40.09
PLEASING	36.31	27.15	40.54	44.96	29.76	19.43	33.93	38.06	33.52	24.17	37.32	42.13
+MM	38.81	29.13	43.07	48.74	31.74	20.88	36.26	40.72	36.26	26.42	40.38	44.26
MM-TKGF	42.34	34.01	46.23	51.50	35.74	26.58	39.72	43.74	39.85	31.57	44.72	49.24

methods (e.g., LogCL), and our conference version TKGF-NTP. Observed from Fig. 7, we have the following findings. (1) As expected, MM-TKGF demonstrates a +36.84% in MRR superiority over static methods. We posit that the noticeable gap between them is due to the static methods neglecting the temporal and multi-modal information of the facts, which generates ambiguous representations of entities that lack multi-modal complementary features and temporal dependencies critical for accurate forecasting. (2) When compared to multi-modal static methods, MM-TKGF achieves a +9.72% MRR improvement. This performance gap arises because they treat MMTKGs as static graphs, thereby ignoring temporal dependencies, which leads to significant performance degradation. To address this, our carefully designed modules AMFE and TGMF enable MM-TKGF to adeptly handle both temporal and multi-modal information. (3) MM-TKGF also yields a +4.95% MRR gain over uni-modal TKG reasoning methods such as LogCL. This superiority stems from our model's ability to effectively integrate entity temporal representation with multi-modal auxiliary information and filter out inter-modal noise. This capability gives MM-TKGF an advantage in multi-modal scenarios, yet these baselines, designed for uni-modal TKG forecasting, cannot leverage multi-modal features, thereby limiting their performance. (4) As anticipated, MM-TKGF outperforms TKGF-NTP by a substantial gain, attaining a +14.62% higher MRR alongside marked improvements across all metrics. This observation indicates that leveraging multi-modal information aids in distinguishing similar entities and enhancing model's performance. To summarize, the superiority of MM-TKGF over existing KG, MMKG, and TKG reasoning methods demonstrates the significance of extending traditional TKG reasoning into a multi-modal scenario.

Moreover, we validate whether existing uni-modal TKG reasoning methods can be enhanced by multi-modal data. To achieve this, we augment these baselines with a fusion module that concatenates the representations derived from baselines with pre-trained

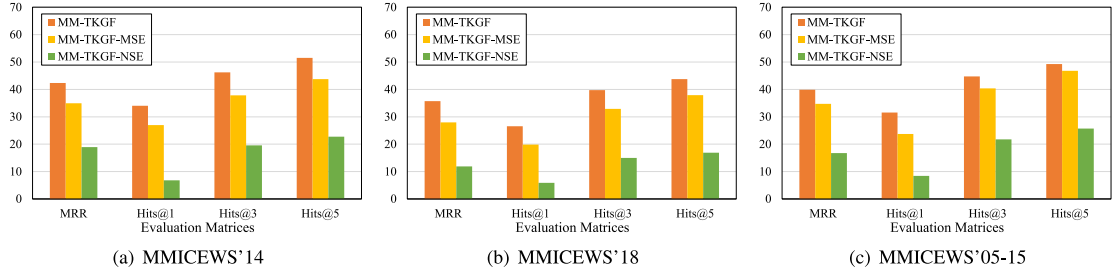


Fig. 8. Impact of different structural encoders.

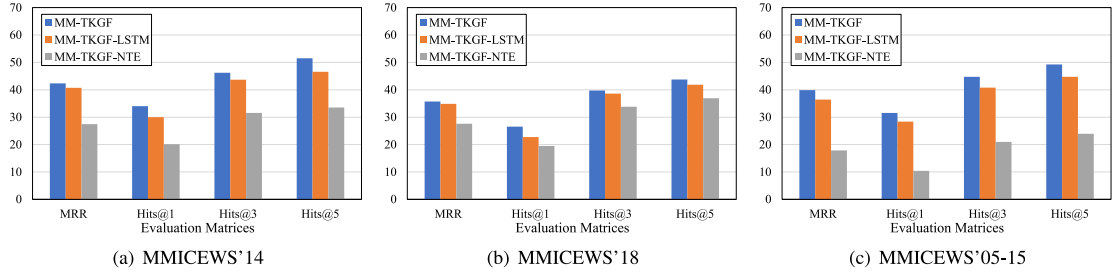


Fig. 9. Impact of different temporal encoders.

multi-modal features, denoted as +MM. As observed from Table 4, integrating multi-modal features improves most uni-modal temporal baselines by a remarkable improvement, with average gains of +3.83%, +6.28%, +2.13%, and +2.32% for MRR, and Hits@1, 3, 5 values, respectively. These improvements confirm that multi-modal data provide complementary features for TKG forecasting methods, and their performance is enhanced by such multi-modal fusion, even through a direct concatenation. Notably, MM-TKGF still outperforms all baselines in most metrics, highlighting the necessity of TGMF module. Unlike direct concatenation, TGMF dynamically fuses multi-modal features and entity temporal representations while filtering out noise, which is essential for accurate forecasting.

5.4. Ablation study (RQ2)

Here, we investigate how different components of MM-TKGF contribute to its effectiveness. Experiments for ablation study are performed across all datasets, with the corresponding results shown in Fig. 8 to 12.

Impact of Different Structural Encoders. To assess the impact of the structural encoder within AMFE module, we maintain the module TGMF, and modify AGNE in various ways as follows. (1) The aggregator, which is implemented in Eq. (7), is substituted with a mean structural encoder, namely MM-TKGF-MSE. (2) AGNE is directly removed (i.e., entities' representations are directly fed into ATE), denoted as MM-TKGF-NSE. We summarize the experimental results in Fig. 8. Most notably, the variant MM-TKGF-NSE performs poorly due to the lack of a structural encoder. This is not surprising, as structural encoder is essential for capturing important graph structural information from each snapshot, removing it will severely impair the model's ability to aggregate valuable structural information from neighbour entities. Meanwhile, the lower performance of MM-TKGF-MSE compared to MM-TKGF implies that treating neighbours equally is ineffective in capturing valuable graph structural information. In other words, for link prediction, assigning attention scores is more effective compared to average weighting to neighbours. In short, the results in Fig. 8 highlight the significant enhancement in model's performance achieved by the attention-based structural encoder, which extracts critical structural information from local snapshots.

Impact of Different Temporal Encoders. Next, we explore the effectiveness of temporal encoder in MM-TKGF. While retaining the module TGMF, the temporal encoder within AMFE is modified in the following ways: (1) ATE is substituted with a recurrent neural network LSTM, referred to as MM-TKGF-LSTM. (2) ATE is directly ablated, labelled as MM-TKGF-NTE. As shown in Fig. 9, without surprise, MM-TKGF consistently outperforms MM-TKGF-LSTM across all metrics. This is because MM-TKGF-LSTM fails to accurately extract the essential temporal information from historical observations. In the variant MM-TKGF-NTE, TGMF module directly integrates structural representations with multi-modal features to produce a unified representation. It is evident that MM-TKGF model surpasses the variant MM-TKGF-NTE across all metrics, due to the absence of a temporal encoder severely limiting the variant's ability to capture temporal dynamics. In conclusion, the results shown in Fig. 9 illustrate that ATE remarkably improves the model's performance, as it adeptly captures the evolution of entities across historical snapshots.

Impact of Different Multi-modal Information. To explore how different types of multi-modal features influence the model MM-TKGF, we keep the module AMFE, and modify the multi-modal data preparation part within TGMF in the following ways. (1)

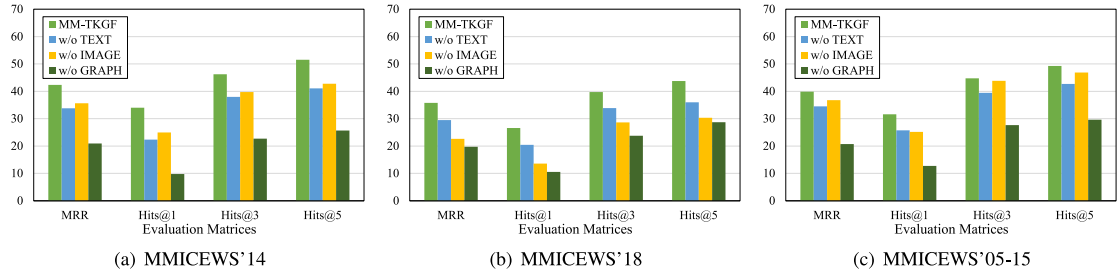


Fig. 10. Impact of different multi-modal information.

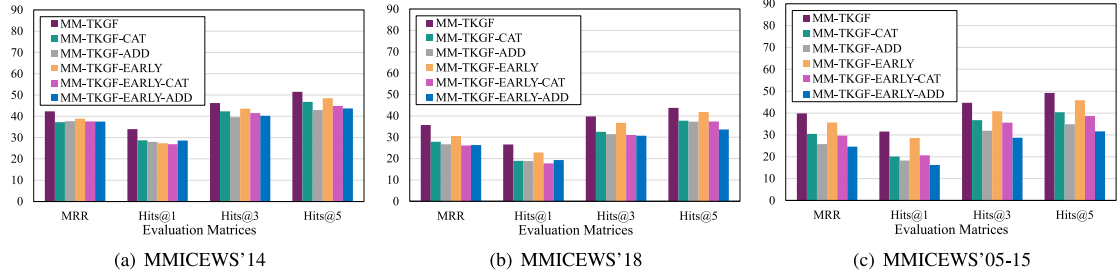


Fig. 11. Impact of different multi-modal fusion strategies.

Textual information is removed, and labelled as w/o TEXT. (2) Image information is not considered (i.e., filling with zero vectors), denoted as w/o IMAGE. (3) A variant in which graph structure is not fed into TGMF, namely w/o GRAPH. The results presented in Fig. 10 indicate that integrating textual, image, and graph structural information leads to a substantially enhancement in all metrics. Reducing any type of information will harm the performance of MM-TKGF to varying degrees. Specifically, the graph structure is the most effective one among all information, and its removal significantly impairs model's performance. It can be naturally understood that when we predict a missing entity for a quadruple, an intuitive solution is to search candidates in MMTKGs based on neighbourhood information rather than multi-modal features.

Impact of Different Multi-modal Fusion Strategies. Furthermore, to assess the impact of multi-modal fusion strategies, we conduct comprehensive experiments within the proposed model MM-TKGF. While keeping the module AMFE, we alter the multi-modal fusion strategy as described below. (1) The gating unit is replaced with a direct concatenation operation, namely MM-TKGF-CAT. (2) We substitute the gating unit with a direct addition operation, denoted as MM-TKGF-ADD. (3) Instead of fusing the multi-modal features with temporal representation of entities, we position the gating unit after AGNE, referred to as MM-TKGF-EARLY. (4) Replacing the gating unit with a direct concatenation method and positioning it after AGNE, labelled MM-TKGF-EARLY-CAT. (5) We modify the gating unit to an add operation, and also position it after AGNE, namely MM-TKGF-EARLY-ADD. As a remarkable observation, the variants with early fusion (i.e., variants (3), (4), and (5)) exhibit lower performance than those with intermediate fusion (i.e., variants (1) and (2)), as shown in Fig. 11. It is revealed that the early fusion may hinder the model's ability to distinguish query-related facts from the global temporal perspective, as discussed in Section 1. The reason behind this phenomenon is that multi-modal features are static, fusing them with structural representations at early stage results in highly similar representations, which impairs the model's performance subsequently. Notably, we also find that MM-TKGF outperforms both MM-TKGF-CAT and MM-TKGF-ADD, indicating that coarse-grained fusion strategies, such as concatenation or addition, impact the performance of MM-TKGF. This is because such strategy cannot filter out the inter-modal noise, and the widespread presence of noise severely harms the model's performance. In a word, the experiment results demonstrate that fusing entities' representations after AMFE module and designing a fine-grained fusion strategy significantly enhance the representations of entities.

Impact of Different Multi-modal Pre-trained Models. To further investigate how the choice of multi-modal pre-trained models affects the performance of MM-TKGF, we keep the module AMFE, and modify the multi-modal feature extraction methods in the following ways. (1) Textual features are pre-trained by BERT (Devlin et al., 2019), and image features are pre-trained by ViT (Dosovitskiy et al., 2021), represented as MM-TKGF-BERT&ViT. (2) Textual features are pre-trained by Doc2vec (Le & Mikolov, 2014), and the image features are pre-trained by ViT, denoted as MM-TKGF-DOC&ViT. (3) Textual features are pre-trained by BERT, and image features are pre-trained by ResNet (He et al., 2016), represented as MM-TKGF-BERT&RES. (4) Textual features are pre-trained by Doc2vec, and image features are pre-trained by VGG (Simonyan & Zisserman, 2015), namely MM-TKGF-DOC&VGG. (5) Textual features are pre-trained by BERT, and image features are pre-trained by VGG, labelled as MM-TKGF-BERT&VGG. (6) Textual and image features are pre-trained by CLIP (Radford et al., 2021), referred to as MM-TKGF-CLIP. Fig. 12 presents the performance of MM-TKGF and its variants that use different multi-modal pre-trained models. Obviously, MM-TKGF obtains the optimal performance across all metrics. This is because the multi-modal features pre-trained by ResNet and Doc2vec provide more complementary features

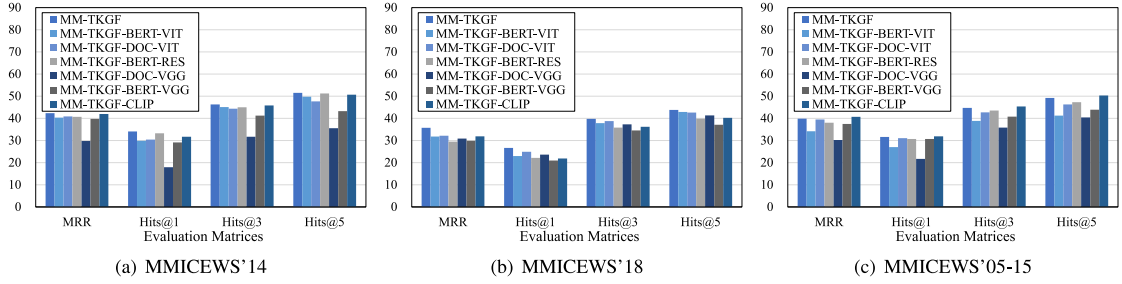


Fig. 12. Impact of different multi-modal pre-trained models.

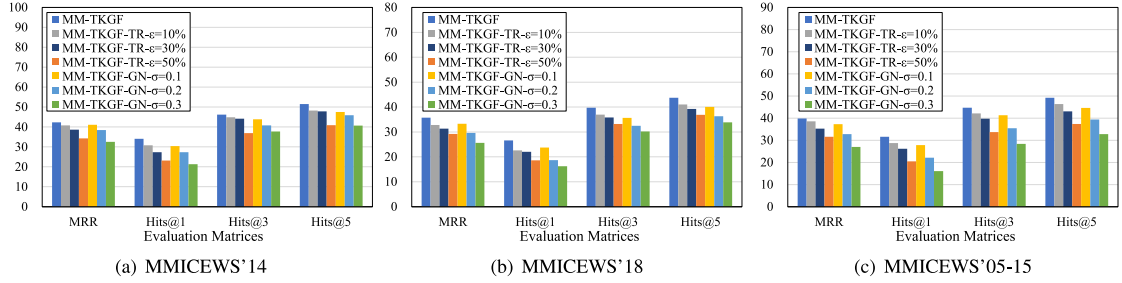


Fig. 13. Impact of different types and levels of multi-modal noise.

to enhance entities' representations. Conversely, other variants exhibit lower performance, possibly due to different combinations of pre-trained models being constrained by their out-of-distribution generalization ability (Naganuma et al., 2023). This limitation hinders the variants from delivering enough complementary features to enhance the representations of entities. To sum up, choosing a suitable pre-trained model can enhance reasoning model's performance.

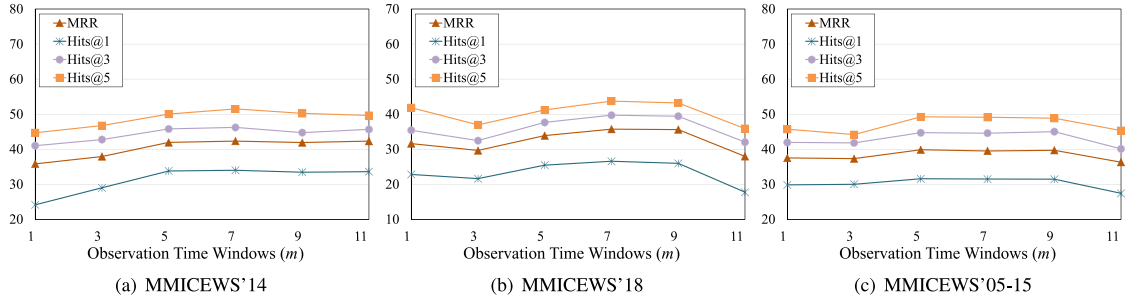
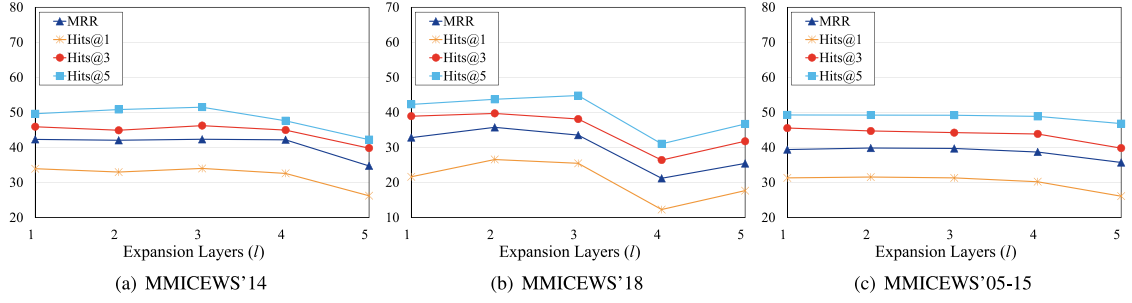
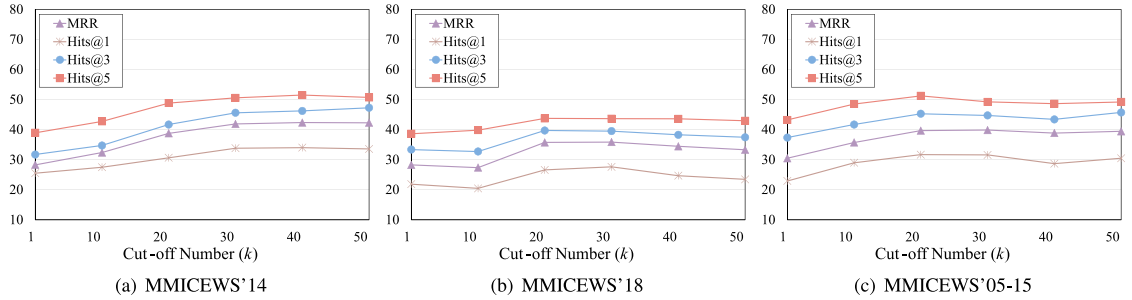
Impact of Different Types and Levels of Multi-modal Noise. To thoroughly evaluate the model's robustness in handling noisy multi-modal information, we conduct extensive experiments by injecting different types and levels of noise into multi-modal information. Specifically, noise is introduced in the following ways. (1) A proportion of tokens (e.g. 10% tokens) in textual information is replaced with random words sampled from the vocabulary. These variants are denoted as MM-TKGF-TR- ϵ , with $\epsilon = 10\%$, 30% , and 50% . (2) Different levels of Gaussian noise ($\mu = 0, \sigma = 0.1, 0.2$, and 0.3) are injected into image information. These variants are referred to as MM-TKGF-GN- σ . The experimental results are summarized in Fig. 13. Interestingly, under mild noise conditions (i.e., $\sigma = 0.1$ for images or $\epsilon = 10\%$ for textual information), the performance of those variants degrades only slightly compared to MM-TKGF. As the noise level increases, the performance decline becomes more noticeable, yet remains within an acceptable margin. Notably, even under severe noise conditions (i.e., $\sigma = 0.3$ for images or $\epsilon = 50\%$ for textual information), our model still outperforms several uni-modal TKG reasoning methods, demonstrating strong robustness against multi-modal noise.

5.5. Hyper-parameter sensitivity analysis (RQ3)

In this part, we systematically investigate the following critical hyper-parameters, i.e., the observation time window m , the number of expansion layers l , the cut-off number k , learning rate η , and the number of snapshots n . A comprehensive hyper-parameter sensitivity analysis is performed across all datasets, with the corresponding results presented from Fig. 14 to 18.

Analysis of Observation Time Window. First, we analyse the parameter observation time window m , which indicates the historical depth of observations to be considered. We perform experiments with m varied from 1 to 11, keeping other hyper-parameters at their optimal values. All metrics, as observed in Fig. 14, present stable and consistent performance gains as the value of m increases. However, setting m to high values may result in performance degradation. This decline is attributed to an excessive value of m leading the model to search for related temporal facts from too far in the past, which complicates the extraction of query-dependent temporal information from such distant observations. This effect is particularly pronounced in datasets with longer time-spans and sparser graph structures (i.e., those with lower ρ values), such as MMICEWS'05-15. There, noisy or out-dated information from distant history can limit the learning of entity embeddings. In contrast, datasets with shorter time-spans and denser graph structures (i.e., higher ρ), like MMICEWS'14 and MMICEWS'18, tend to benefit from a larger m value, as it helps capture more relevant recent context within the denser TKG snapshot. To achieve optimal performance, we set m to 7, 7, and 5 on datasets MMICEWS'14, MMICEWS'18, and MMICEWS'05-15, respectively.

Analysis of Expansion Layer. Next, we analyse the extent to which the number of expansion layers l affects the model's performance. Varying l from 1 to 5 while fixing other hyper-parameters to their optimal values. As shown in Fig. 15, 2-layers

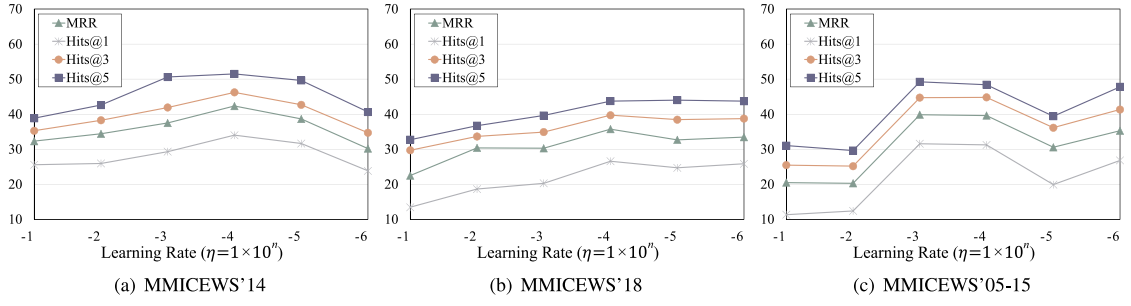
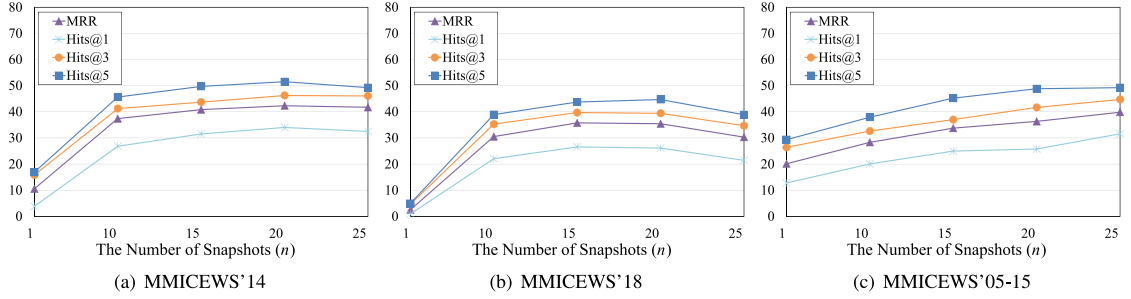
Fig. 14. Comparison of different observation time window (m)Fig. 15. Comparison of different expansion layers (l) for structural encoder.Fig. 16. Comparison of various cut-off numbers (k)

structural encoder outperforms the 1-layer one. This improvement is due to the 2-layers encoder aggregating multi-hop neighbours that provide more structural information than the 1-hop neighbour. Besides, our observations indicate that MM-TKGF performs best with either 2 or 3 layers. Yet, once l exceeds three, the metrics tend to decline. This is because although additional layers allow the encoder to aggregate further structural information, excessive layers might lead to over-fitting. Finally, the optimal setting for l is 3, 2, and 2 on datasets MMICEWS'14, MMICEWS'18, and MMICEWS'05-15, respectively.

Analysis of Cut-off Number. To further investigate how the cut-off number k affects the performance of MM-TKGF. The value of k is set to 1, and 10 to 50 in steps of 10, while other hyper-parameters are optimized. The results, which initially show an upward trend, are depicted in Fig. 16. As expected, setting k to 1 severely impairs the performance of MM-TKGF, as considering only one neighbour entity leads to inadequate feature extraction. Furthermore, as k increases significantly, considering too many neighbours increases running time to unacceptable levels and introduces unexpected noise into the representations. For optimal performance, k is assigned to 40, 20, and 30 on MMICEWS'14, MMICEWS'18, and MMICEWS'05-15, respectively.

Analysis of Learning Rate. Besides, the influence of learning rate η is explored by conducting a series of experiments with different η values from $\eta = 1 \times 10^{-6}$ to 1×10^{-1} , while other hyper-parameters are fixed. As revealed in Fig. 17, the experimental results first exhibit low performance with a very small learning rate, likely due to the model becoming trapped in local minima. The performance then shows a stable upward trend with an increase of η . However, once the learning rate reaches 1×10^{-2} , the evaluation metrics start to decline, due to overly large learning rates causing the model to overshoot the optima. The optimal learning rates are determined to be 1×10^{-4} , 1×10^{-4} , and 1×10^{-3} on MMICEWS'14, MMICEWS'18, and MMICEWS'05-15, respectively.

Analysis of Snapshot Number. Additionally, we assess the impact of the number of snapshots n , which indicates how many snapshots are used to construct the MMTKG. While other hyper-parameters are kept, n is varied from 1 to 25. Notably, when $n = 1$,

Fig. 17. Comparison of various learning rates (η)Fig. 18. Comparison of various numbers of snapshots (n)

the dataset comprises a single snapshot, i.e., the MMTKG degrades into the static MMKG. For n values of 5, 10, 15, 20, or 25, the set is partitioned into 5, 10, 15, 20, or 25 snapshots, respectively, ordered chronologically. As illustrated in Fig. 18, there is an initial upward trend in model's performance with increasing n . This increase is attributed to the fact that ATE enables the model to utilize information across multiple snapshots, thereby enhancing its ability to capture temporal dynamics among entities. However, an excessive number of snapshots may limit the structural encoder's capacity to extract valuable information from sparse MMTKG snapshots. Overall, to achieve optimal performance, the snapshot numbers are set to 20 for MMICEWS'14, 15 for MMICEWS'18, and 25 for MMICEWS'05-15, respectively.

5.6. Qualitative study (RQ4)

In this section, we qualitatively elucidate the effectiveness of MM-TKGF through predication case studies and cross-modal embedding visualizations, revealing interpretable insights for model performance.

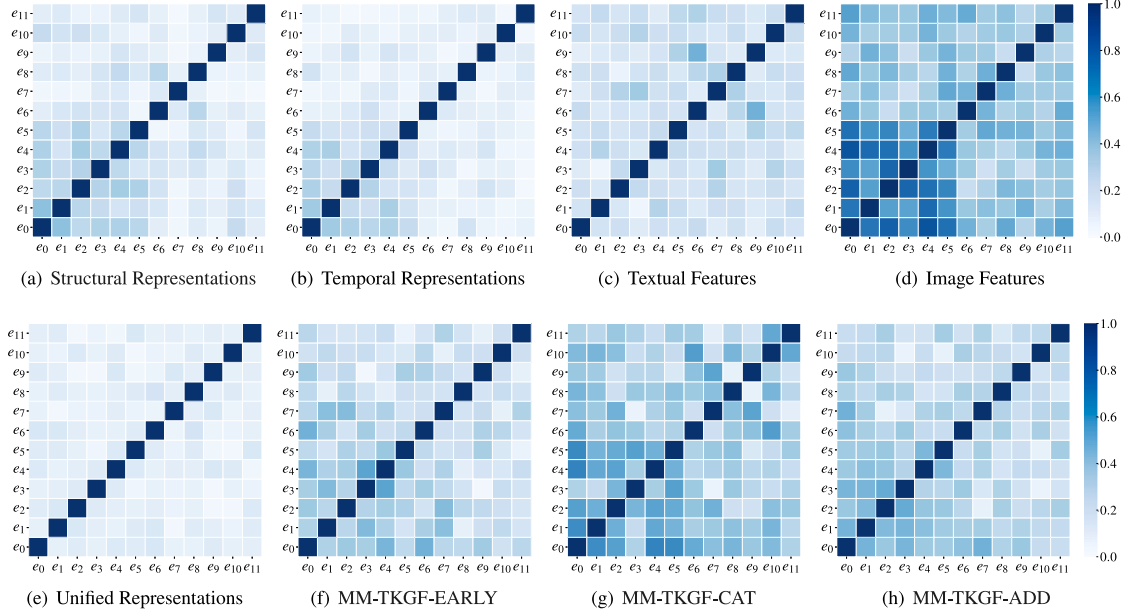
Case Analysis. To further illustrate the importance of each key module of MM-TKGF, Table 5 reports the top-3 prediction scores for three query cases across different MMTKG datasets (i.e., MMICEWS'14, MMICEWS'18, and MMICEWS'05-15), comparing MM-TKGF against its ablated variants (MM-TKGF-NSE, MM-TKGF-NTE, and w/o GRAPH). As shown, MM-TKGF correctly identifies the answer with high confidence in all cases. In contrast, both MM-TKGF-NSE and w/o GRAPH consistently fail to include the correct answer among their top-3 predictions across all three queries, due to fundamental architectural limitations. Specifically, MM-TKGF-NSE lacks the ability to capture local structural dependencies as it omits structural encoder, while w/o GRAPH relies solely on static multi-modal features that lack both structural and temporal dependencies, severely hindering its reasoning ability. Notably, MM-TKGF-NTE includes the correct answer for Query1 and Query2 in its top-3 predictions but assigns it a lower confidence than MM-TKGF. This occurs because MM-TKGF-NTE depends primarily on local structural dependencies and static multi-modal features while neglecting the temporal dynamics of entities. These specific cases across diverse datasets validate the effectiveness and generalization of MM-TKGF's core modules and highlight its superior reasoning ability.

Visualization of Cross-modal Embeddings. To demonstrate MM-TKGF's ability to distinguish similar entities, Fig. 19 presents a case study visualizing pair-wise similarities for twelve MMICEWS'14 entities. Specifically, the sampled entities comprise two groups. Politicians $e_0 \sim e_5$ inherently share highly similar underlying graph structures. However, their multi-modal features typically include photographs, signature imagery, and biographic attribute texts, exhibiting substantial diversity due to personalized attributes and thereby contrasting with their similar graph structure. In comparison, entities $e_6 \sim e_{11}$ respectively represent an organization, a state, a group, an army, a university, and a corporation. These entities are characterized by landmarks, logotypes, or descriptions that directly reflect their functional nature. Fig. 19 (a) to (e) depict embedding similarities derived from different MM-TKGF components. Among them, heat maps (a) and (b) respectively visualize structural $\mathcal{X}_{e,t}$ and temporal $\mathcal{H}_{e,t}$ representations from AMFE module, while (c) and (d) present text-based $\mathcal{M}_e^t$ and image-based $\mathcal{M}_e^i$ features, respectively. The unified representations $\mathcal{Z}_{e,t}$ produced by TGME

Table 5

Top-3 predictions of MM-TKGF variants for exemplary query cases.

Query and Answer	MM-TKGF		MM-TKGF-NSE		MM-TKGF-NTE		w/o GRAPH	
	Candidate	Score	Candidate	Score	Candidate	Score	Candidate	Score
Query1: (<i>John Kerry, Discuss by telephone, ?, 2014-12-22</i>)	Wang Yi	0.62	Sergey Lavrov	0.31	Sergey Lavrov	0.53	Moldova	0.09
	Sergey Lavrov	0.16	Mahmoud Abbas	0.14	Benjamin Netanyahu	0.11	United Nations	0.09
Answer: Wang Yi	Laurent Fabius	0.13	United States	0.11	Wang Yi	0.09	The Hague	0.09
Query2: (<i>Theresa May, Make a visit, ?, 2018-12-13</i>)	Belgium	0.55	United Kingdom	0.23	Angela Merkel	0.30	Somalia	0.10
	European Union	0.13	Donald Tusk	0.12	Belgium	0.14	Michel Barnier	0.10
Answer: Belgium	United Kingdom	0.10	European Union	0.11	United Kingdom	0.08	Jorge Arreaza	0.07
Query3: (<i>Google, Express intent to cooperate, ?, 2015-12-18</i>)	Facebook	0.52	United States	0.15	European Commission	0.24	Milos Zeman	0.08
	United States	0.09	Singapore	0.11	United States	0.13	France	0.08
Answer: Facebook	South Korea	0.07	South Korea	0.10	Barack Obama	0.09	Tony Abbott	0.08

**Fig. 19.** Heat maps of cross-modal entity embeddings.

module are showcased in (e). As shown in (a), there is a high similarity between different types of entities, such as e_6 and e_8 . This indicates a limitation that static structural representations cannot capture the evolving patterns of entities, which may explain why different types of entities appear similar from a local perspective. To address this, temporal representations can effectively model such temporal dynamics, thereby representing entity semantics more accurately from a global perspective. Notably, both (a) and (b) exhibit high similarity among politicians $e_0 \sim e_5$, contrasting sharply with those of the other entities $e_6 \sim e_{11}$. This phenomenon demonstrates an inherent constraint of (a) and (b), as they cannot distinguish between entities that have similar graph structures. In contrast, the textual (c) and image (d) modalities provide complementary features for these politicians, offering additional perspectives beyond the shared graph structure. Crucially, the fused representations in (d) achieve pronounced differentiation by successfully resolving fine-grained distinctions among the highly similar politicians, while clearly distinguishing them from other entities $e_6 \sim e_{11}$, thereby overcoming the limitations observed in uni-modal representations.

Visualization of Fusion Strategies. To further validate the design of the TGMF module, we provide a visual analysis of the pair-wise similarity based on the unified representations of different ablated variants. As an extension of heat maps (a) ~ (e), (f) ~ (h) illustrate similarity patterns under several typical fusion strategies. Specifically, heat map (e) corresponds to the unified representations from our complete MM-TKGF. Heat map (f) shows MM-TKGF-EARLY, which fuses multi-modal features before ATE module. The last two heat maps present gating-free variants: MM-TKGF-CAT (g) directly concatenates multi-modal features with entity temporal representations, while MM-TKGF-ADD (h) fuses them via addition. As clearly evident from heat map (e), the gating mechanism in our TGMF module demonstrates a superior ability to differentiate similar entities $e_0 \sim e_5$, as shown by the distinctly lower pair-wise similarities among them. Through the vivid visualization in heat map (f), we can observe that early fusion strategy discussed previously (Section 5.4) leads to higher vector similarity in entity representations, which strongly support our design to position TGMF module after ATE component. The ablation variants with simpler operations (g) and (h), however, not only fail to distinguish these similar entities but also introduce abnormally higher similarity among other entities with different semantics, such

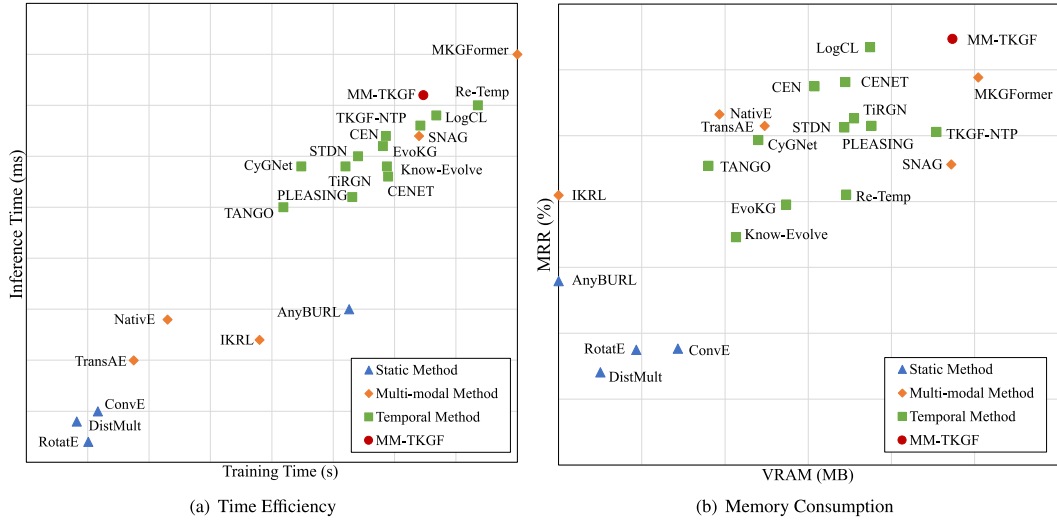


Fig. 20. Comparison on efficiency and memory cost.

as e_6 (an organization) and e_{10} (a university). This clear divergence revealed in the heat maps validates the critical role of our TGMF module in enhancing entity representation quality, particularly in distinguishing similar entities.

5.7. Efficiency study (RQ5)

This section formally evaluates the efficiency of MM-TKGF through complexity analysis and efficiency comparison, demonstrating the practical value of the proposed model.

Theoretical Complexity Analysis. The computation overhead of MM-TKGF primarily arises from its core modules AMFE and TGMF. To be specific, the time complexity of AMFE is dominated by two key components: the query-dependent attention aggregator in AGNE with a complexity of $\mathcal{O}(|\mathcal{E}|kd^2 + n_md)$ and the self-attention layer in ATE with a complexity of $\mathcal{O}(|\mathcal{E}|md^2)$, where n_m denotes the maximum number of neighbours. Consequently, the overall complexity of AMFE is $\mathcal{O}(|\mathcal{E}|(k + \frac{m}{l})d^2 + n_md)$. When combined with the complexity of TGMF $\mathcal{O}(|\mathcal{E}|d(d_i + d_l + d))$, this yields the total time complexity of MM-TKGF, i.e., $\mathcal{O}(|\mathcal{E}|d(k + \frac{m+1}{l})d + d_i + d_l + n_m)$.

Empirical Efficiency Validation. To further validate the computational efficiency of MM-TKGF, we compare it with all baselines under identical hardware configurations. As shown in Fig. 20 (a), our model achieves a balanced trade-off between inference and training time. In terms of memory consumption (Fig. 20 (b)), MM-TKGF delivers superior MRR performance without excessive GPU memory usage, underscoring its effective resource-performance balance. While some uni-modal baselines may be marginally faster or consume less memory, they suffer from a significant cost to MRR performance. Overall, MM-TKGF achieves an optimal trade-off among time efficiency, resource consumption, and performance, yielding the best results while maintaining competitive training speeds and modest resource demands. Although the multi-modal fusion module TGMF incurs computational overhead to some extent, this cost remains acceptable, considering the significant performance gains it delivers.

6. Implication

Theoretical Implications. We extend the boundaries of TKG forecasting task by explicitly incorporating multi-modal features. By formalizing this novel task, we synthesize two disjoint tasks: MMKG reasoning and TKG forecasting through a unified problem definition, paving the way for future research in MMTKG reasoning.

Practical Implications. We propose the first forecasting model for MMTKG, which jointly leverages entity temporal representations and multi-modal features for accurate forecasting. Beyond technological innovation, we validate that integrating multi-modal features can improve uni-modal TKG models, revealing the untapped potential of multi-modal data in TKG reasoning. These advances directly empower downstream applications like event prediction, where temporal patterns and multi-modal data co-exist.

7. Conclusion

Knowledge graphs are vital for numerous downstream applications, including online recommendation, question answering, and social network analysis. To further enhance existing methods' performance on TKG forecasting, the task is extended to embrace multi-modality and a brand new model, namely MM-TKGF, is designed. In detail, the model comprises two modules, i.e., AMFE and TGMF. Specifically, in the former, we design a group of encoders called Attention-based Multi-faceted Encoder. The module aims to mine the graph structure to capture the entities' structural dependencies and temporal evolution. In the latter, we develop a

series of multi-modal fusion units called Temporal-aware Gated Multi-modal Fusion Module. This module is dedicated to leveraging the multi-modal features effectively and generating fine-grained complementary representations from multi-source information. To assess the effectiveness of our proposed MM-TKGF, exhaustive experiments are performed across three datasets, demonstrating its superiority over existing methods. In future, we will focus on more challenging scenarios, such as reasoning over disconnected, extremely large scale, or incremental multi-modal temporal knowledge graphs (Xie et al., 2024).

CRedit authorship contribution statement

Gaojie Han: Writing – original draft, Methodology, Data curation. **Wei Chen:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Xiaofang Zhang:** Writing – review & editing, Supervision, Formal analysis. **Li Zhang:** Writing – review & editing, Validation, Methodology. **An Liu:** Writing – review & editing, Methodology, Formal analysis. **Lei Zhao:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

All authors disclosed no relevant relationships.

Data availability

Data will be made available on request.

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